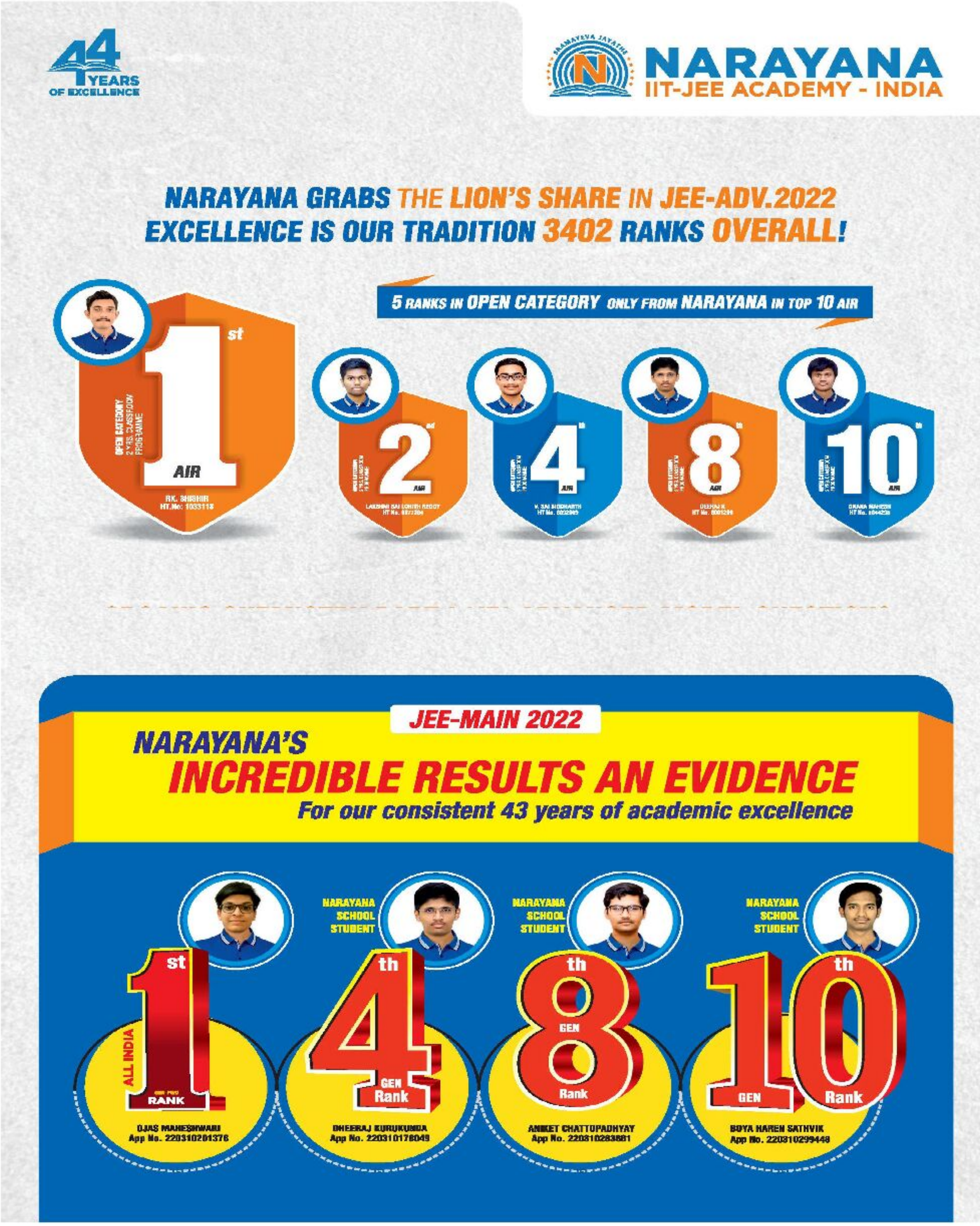


MATHS

CO ORDINATE GEOMETRY
JEE ADVANCED IMPORTANT QUESTIONS

SINGLE CORRECT TYPE QUESTION

1. If line through $P(a, 2)$ meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at A and D and also meets the axes at B and C, so that PA, PB, PC, PD are in G.P., then

A) $a\in[5, \infty)$ B) $a\in R$ C) $a\in(-\infty, -6)\cup[6, \infty)$ D) none of these

2. The length of focal chord AB of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ is: (Given $A=\left(\frac{8}{5}, \frac{3\sqrt{3}}{5}\right)$)

(A) $\frac{4}{5}$ (B) $\frac{16}{5}$ (C) $\frac{32}{5}$ (D) $\frac{64}{5}$

3. In Δ ABC, $2x + 3y + 1 = 0$, $x + y = 1$ and $2x - y + 1 = 0$ are median through vertex A, altitude through vertex B and angle bisector of C respectively

(A) intercept made by BC on y-axis is $\frac{379}{59}$

(B) area of Δ ABC is $\frac{401}{59}$

(C) intercept made by BC on x-axis is $\frac{379}{423}$

(D) none of these

4. Image of the curve $xy=1$ in the curve $\left(x+\sqrt{x^2+1}\right)\left(y+\sqrt{y^2+1}\right)=1$ (on coordinate plane) is

(A) $xy=1$ (B) $xy+1=0$ (C) $xy=0$ (D) $x^2+y^2=1$

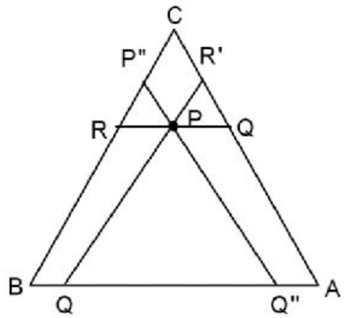
5. Consider two parabolas: $y=x^2-x+1$ *and* $y=-x^2+x+\frac{1}{2}$, the parabola $y=-x^2+x+\frac{1}{2}$ is fixed and parabola $y=x^2-x+1$ rolls without slipping around the fixed parabola, then the locus of the focus of the moving parabola is

(A) $y=1$ (B) $y=\frac{3}{4}$ (C) $y=x^2$ (D) $y=-x^2$

6. Three circles C_1, C_2, C_3 with radii r_1, r_2, r_3 ($r_1< r_2< r_3$) respectively are given as $r_1 = 2$, and $r_3 = 8$ they are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. There exist two straight lines each of which is a direct common tangent simultaneously to all the three circles then r_2 is equal to

(A) $r_2=4$ (B) $r_2=5$ (C) $r_2=10$ (D) $r_2=16$

7. In a Δ ABC, AB = 85, BC = 90 & AC =102. P is an interior point & line segments are drawn through P parallel to the sides of the triangle. If these three line-segments (as shown in the diagram by RQ, R'Q', R''Q'') are of equal length d, then $\frac{5d}{34}$ is equal to



(A) 9 (B) 6 (C) 3 (D) 12

8. Let A and B be the ends of the diameter $4x - y - 15 = 0$ of the circle $x^2 + y^2 - 6x + 6y - 16 = 0$. A and B lie on tangents at the end points of the major axis of an ellipse such that line joining A and B is a tangent to the same ellipse at a point P. If the equation of major axis of the ellipse is $y = x$ then the distance between the foci is

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 8 (D) $2\sqrt{3}$

9. Let ABC be a right-angle triangle right angled at A, S be its circumcircle. Let S_1 be the circle touching the lines AB and AC and the circle S internally. Let S_2 be the circle touching the lines AB, AC and the circle S externally. If r_1, r_2 be the radii of the circle S_1 and S_2 respectively then area Δ ABC is

A) $\frac{r_1r_2}{4}$ B) $\frac{r_1r_2}{8}$ C) $\frac{r_1r_2}{2}$ D) r_1r_2

10. If tangents are drawn to the parabola $y = x^2 + bx + c$ for b and c fixed real number at the points (i, y_i) for $i = 1, 2, \ldots, 10$. Let $I_1, I_2, I_3 \ldots I_9$ be the point of intersection of tangents at (i, y_i) and $(i + 1, y_{i+1})$ then the least polynomial satisfying whose graph passes through all nine points is,

(A) $y = x^2 + bx + c$ (B) $y = x^2 + bx + c - \frac{1}{2}$

(C) $y = x^2 + bx + c - \frac{1}{4}$ (D) $y = x^2 + bx + c - \frac{1}{8}$

11. In triangle ABC, if $y = x+1$ & $y = 2x + 3$ are altitude through A & angle bisector of B respectively. If vertex $C \equiv (3 , 4)$ then equation of median through C is

(A) $x + 7y = 25$ (B) $7x + 2y = 25$

(C) $7x + y = 25$ (D) $2x + 7y = 25$

12. If the line $x + y + 1= 0$ and $y - 2x + 5 = 0$ are tangents to a parabola whose focus is at (1, 2), then the equation of normal to the parabola through $\left(\frac{29}{17}, -\frac{14}{17}\right)$ is,

(A) $2x + 4y - 1= 0$ (B) $y - 3x + 2 = 0$

(C) $4x + y - 5 = 0$ (D) None of these

13. For a parabola $x^2 - 4xy + 4y^2 - 32x + 4y + 16 = 0$, focus is

A) $(2, 1)$ B) $(-2, 1)$ C) $(-2, -1)$ D) $(2, -1)$

14. If $2x - y + 1 = 0$ is a tangent on the parabola which intersect its directrix at (1, 3) and focus is (2, 1), then equation of axis of parabola is
(A) $11x - 2y = 24$ (B) $11x + 2y + 24 = 0$
(C) $2x - 11y - 24 = 0$ (D) $2y + 11x = 24$

15. A parabola having directrix $2x + y = 3$ touches a line $x + y = 2$ at $(3, -1)$. If focus of parabola is (α, β), then $\alpha + 2\beta$ is equal to
(A) 1 (B) 2 (C) 3 (D) 4

16. A circle C_1 with radius 5 touches x-axis and another circle C_2 with radius 4 touches y-axis. The two circles touches each other externally so that their point of contact lies in the first quadrant, let the locus of their point of contact be the curve S. Two tangents are drawn to the curve S from point P(9, 5) to meet the curve at Q and R. The area of ΔPQR is,
A) $\frac{29}{4}$ B) $\frac{27}{5}$ C) $\frac{13}{2}$ D) $\frac{25}{3}$

MULTI CORRECT QUESTIONS

17. For the ellipse $12x^2 - 12xy + 7y^2 = 48$
(A) slope of major axis is $\frac{3}{2}$ and length$= 8$
(B) slope of minor axis is $\frac{-2}{3}$ and length$= 6$
(C) coordinate of foci are $(2,3)$ and $(-2,-3)$
(D) Equation of directrices are $2x+3y+14=0$ and $2x+3y-14=0$

18. A point M divides A and B in the ratio 1 : 2 where A and B are diametrically opposite ends of a circle
$x^2 + y^2 - 5x - 9y + 22 = 0$. Square AMCD and BMEF on the length AM and MB are constructed on the same side of line AB. If co-ordinates of A is (1, 3) then the orthocenter of Δ ABE is,
(A) (1, 6) (B) (1, 5) (C) (3, 3) (D) (4, 6)

19. Normals are drawn from the external point (h, k) to the rectangular hyperbola $xy = c^2$. If circles are drawn through the feet of these normals taken three at a time then centre of circle lies on another hyperbola for which centre and eccentricity is,
A) $Center \equiv \left(\frac{h}{2}, \frac{k}{2}\right)$ B) $Center \equiv (h, k)$ C) $e = \sqrt{2}$ D) $e = \sqrt{2} + 1$

20. Minimum value of $\sqrt{2x^2 + 2x + 1} + \sqrt{2x^2 - 10x + 13}$ is $\sqrt{\alpha}$, then
A) α is an even number B) sum of digits of α is 2
C) number of prime factors of α is 2 D) number of prime factors of α is 3

21. Let x – 2y – 5 = 0 be the directrix of a parabola and x – y – 7 = 0 be the tangent drawn to the parabola at the point P(4, –3), then which of the following are **INCORRECT?**

(A) length of latus rectum = $\frac{16}{\sqrt{5}}$

(B) harmonic mean of length of segments of the focal chord = $\frac{8}{\sqrt{5}}$

(C) intersection point of the directrix and the given tangent is (9, 2)

(D) circle drawn on assuming P(4, –3) and Q as diameter, always passes through focus of the parabola (where Q is the intersection point of tangent and directrix)

22. Let ABCD be a quadrilateral inscribed in a circle. Suppose AB = $\sqrt{2+\sqrt{2}}$ and AB subtends 135° at the centre of the circle, the maximum possible area of ABCD is

A) $\frac{5+3\sqrt{3}}{4\sqrt{2}}$ B) $\frac{8+\sqrt{7}}{\sqrt{3}}$ C) $\frac{19+\sqrt{3}}{2}$ D) none of these

23. x + y = 2 and x – y = 2 are tangents on a parabola at (1, 1) and (4, 2) respectively. Which of the followings is/are correct.

(A) Equation of directrix is x + 3y = 2 (B) Equation of axis is 3x – y = 5

(C) Focus of the parabola is at $\left(\frac{8}{5},\frac{6}{5}\right)$ (D) Vertex of the parabola is at $\left(\frac{33}{20},\frac{13}{20}\right)$

24. If A_i is the area bounded by $|x-a_i|+|y|=b_i,\ i\in N$, where $a_{i+1}=a_i+\frac{3}{2}b_i$ & $b_{i+1}=\frac{b_i}{2}, a_1=0, b_1=32$, then

A) $A_3=128$ B) $A_3=256$

C) $\lim_{n\to\infty}\sum_{i=1}^{n}A_i=\frac{8}{3}(32)^2$ D) $\lim_{n\to\infty}\sum_{i=1}^{n}A_i=\frac{4}{3}(16)^2$

25. The line L is drawn through the point (1, 1) to intersect L_1 : x + 2y = 1 and L_2 : x + 2y = 3 at P and Q respectively. Line perpendicular to L from P intersects L_2 at R and line perpendicular to L from Q intersects L_1 and S. If the area of parallelogram PQRS is least, then which of the following is/are correct?

(A) the equation of line L can be y + 3x = 4 (B) the equation of line L can be 3y – x = 2

(C) the least area of PQRS is $\frac{8}{5}$ square units (D) the equation of line L can be 3y – x = 4

26. Tangents are drawn to parabola $y^2 = 16x$ at the point's A, B and C such that three tangents form a triangle PQR. If θ_1, θ_2 and θ_3 be the inclinations of these tangents with the axis of x such that

their cotangents form an A.P. with common difference 3. Then which of following are correct:

(A) Area of ΔPQR is 432
(B) Area of ΔABC is 832
(C) Area of ΔPQR is 416
(D) Area of ΔABC is 864

27. A parabola $S = 0$ has its vertex at (-9, 3) and it touches the x-axis at the origin then equation of axis of symmetry of the aforesaid parabola can be.

(A) $x - y + 12 = 0$
(B) $x - 2y + 15 = 0$
(C) $2x - y + 21 = 0$
(D) $x + y + 6 = 0$

28. The tangent to the circle $x^2 + y^2 = 4$ at a point P intersect the parabola $y^2 = 4x$ at points Q and R. Tangents to the parabola at Q and R intersect at S. If Q lies in the first quadrant such that PQ = 1 unit, then

(A) PR = 8 units
(B) $SR = \frac{35}{4}$ units
(C) $SQ = \frac{7}{2}$ units
(D) area of $\Delta SQR = \frac{343}{16}$ sq. units

29. A parabola touches x-axis at origin such that foot of perpendicular from origin to axis of parabola is N(-2, 2) then

(A) eqn. of directrix is $x + y + 4 = 0$
(B) Focus of parabola is (-2, 2)
(C) Length of latus rectum is $2\sqrt{2}$
(D) eqn. of axis of parabola is $y = x + 4$

30. If P is any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose foci are S_1 & S_2. Let $\angle PS_1S_2 = \alpha$ & $\angle PS_2S_1 = \beta$ then,

A) $PS_1 + PS_2 = 2a$, if $a > b$

B) $PS_1 + PS_2 = 2b$, if $a < b$

C) $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2} = \frac{1-e}{1+e}$

D) $\tan\frac{\alpha}{2}.\tan\frac{\beta}{2} = \frac{\sqrt{a^2 - b^2}}{b^2}\left(a - \sqrt{a^2 - b^2}\right)$

31. Tangent drawn at point P(1, 3) of a parabola intersects its tangent at vertex at M(−1, 5) and cuts the axis of parabola at T. If R(−5, 5) is a point on SP; where S is focus of the parabola, then which among the following option(s) is/are correct ?

(A) Slope of axis is −3

(B) Radius of circumcircle of ΔSMP is $\sqrt{\frac{5}{2}}$ units

(C) $(ST)^2 - (SM)^2 - (PM)^2$

(D) Tangent cuts the axis of parabola at T (−3, 7)

32. Let ABCD be a rectangle with A(0, 0), B(4, 0), C(4, 4) and D(0, 4). Rectangle is folded in such a way that corner B always lies on line AD, then which of the following statement(s) is/are correct?

(A) As the point B moves on AD the crease thus formed will touch a fixed parabola whose focus is at (4, 0)

(B) As the point B moves on AD the crease thus formed will touch a fixed parabola whose focus is at $\left(\frac{3}{2}, 0\right)$

(C) As the point B moves on AD the crease thus formed will touch a fixed parabola whose equation is $y^2 = 8(x-2)$

(D) As the point B moves on AD the crease thus formed will touch a fixed parabola whose equation of directrix is $y = 2x$

33. Let A, B, C and D be four distinct point on a line in that order. The circles with diameters AC is $x^2 + y^2 + ax + c = 0$ and BD is $x^2 + y^2 - by = 0$ intersect at X and Y,the line XY meets BC at Z. Let P be a point on XY other than Z, the line CP intersects the circle with diameter AC at C and M, and line BP intersects the circle with diameter BD at B and N and the equation of line AM and DN are $bx + cy + a = 0$ and $cx + ay + b = 0$ respectively, then which of the following is true (where ω is a cube root of unity)

(A) $a+b+c=1$ (B) $a\omega+b\omega^2+c=0$

(C) $a+b\omega^2+c\omega=0$ (D) none of these

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 34 and 35

Let $S=0$ be the equation of the locus of point P in the plane of a square $ABCD$ such that $\max(PA,PC)=\frac{1}{\sqrt{2}}(PB+PD)$

34. The ratio of radii of circumcircle of square ABCD and director circle of $S=0$ is,

A) $\frac{\sqrt{3}}{2}$ B) $\frac{1}{4}$ C) $\frac{1}{2}$ D) None of these

35. The ratio of perimeter of S=0 and the given square is

A) $\frac{1}{2\sqrt{2}}$ B) $\frac{1}{\sqrt{3}}$ C) $\frac{1}{7}$ D) $\frac{\pi}{2\sqrt{2}}$

Paragraph for Question Nos. 36 and 37

The centres of two circles C_1 and C_2 of radii 4 and 9 respectively are at a distance of 15 units from each other. Let L_1 be one direct common tangent to these circles.

36. There are n circles which touches both the circle C_1 and C_2 and also the line L_1, then n is

(A) 2 (B) 4 (C) 3 (D) 8

37. Let $r_1, r_2, ..., r_n$ be the radii of these circles then find max $(r_1, r_2, ..., r_n)$?

(A) $\frac{25}{2}$ (B) $\frac{50}{9}$ (C) 50 (D) 1

Paragraph for Question Nos. 38 and 39

Read the following write up carefully and answer the following questions:

A variable line L intersects the parabola $y = x^2$ at points P and Q whose x-coordinate are α and β respectively with $\alpha < \beta$ the area of the figure enclosed by the segment PQ and the parabola is always

equal to $\frac{4}{3}$. The variable segment PQ has its middle point as M

38. Which of the following is/are correct?

(A) $(\beta - \alpha)$ can have more than one real values

(B) $(\beta - \alpha)$ can be equal to 2

(C) $(\beta - \alpha)$ can have exactly one real value

(D) $\alpha = 2 + \beta$

39. Which of the following is/are correct?

(A) equations of the pair of tangents, drawn to the curve, represented by locus of M from origin are $y = 2x$ and $y = -2x$

(B) equation of pair of tangents to be curve, represented by locus of M from origin are $y = x$ and $y = -x$

(C) area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it from origin is $\frac{2}{3}$ sq. units

(D) area of the region enclosed between the curve, represented by locus of M, and the pair of tangents drawn to it, from origin is $\frac{1}{3}$ sq. units

Paragraph for Question Nos. 40 and 41

Read the following write up carefully and answer the following questions:

Diameter of ellipse is a line passing through the centre of the ellipse. Consider the set of parabola(s) having common chord of minimum length with the diameter of the ellipse $x^2 + y^2 + xy = 8$ and the line $mx - y + 4 = 0$ as directrix.

40. The set of value(s) of m for which only one parabola is possible is

A) $\{-\sqrt{2}, \sqrt{2}\}$ B) $\{-\sqrt{5}, \sqrt{3}\}$ C) $\{-2, 2\}$ D) none of these

41. The range of value(s) of m for which two parabola(s) are possible is

A) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ B) $(-\infty, -2) \cup (2, \infty)$

C) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$ D) none of these

Paragraph for Question Nos. 42 and 43

Let ABCD is a rectangle with AB = a & BC = b & circle is drawn passing through A & B and touching side CD. Another circle is drawn passing though B & C and touching side AD. Let r_1 & r_2 be the radii of these two circle respectively

42. $\frac{r_1}{r_2}$ equals

A) $\frac{a}{b}\left(\frac{4b^2+a^2}{4a^2+b^2}\right)$ B) $\frac{b}{a}\left(\frac{4a^2+b^2}{4b^2+a^2}\right)$ C) $\frac{a}{b}\left(\frac{4b^2-a^2}{4a^2-b^2}\right)$ D) $\frac{b}{a}\left(\frac{a^2-4b^2}{4a^2-b^2}\right)$

43. Minimum value of $(r_1 + r_2)$ equals

A) $\frac{5}{8}(a-b)$ B) $\frac{5}{8}(a+b)$ C) $\frac{3}{8}(a-b)$ D) $\frac{3}{8}(a+b)$

Paragraph for Question Nos. 44 and 45

A parabola passes through the points A and B, the ends of a diameter of a given circle of radius 'a'. A tangent to the concentric circle of radius 'b' is the directrix of the parabola.

44. If AB and a perpendicular diameter are taken as coordinate axes, the locus of the focus of the parabola is

A) $\frac{x^2}{a^2}+\frac{y^2}{b^2-a^2}=1$ B) $\frac{x^2}{a^2}-\frac{y^2}{b^2-a^2}=1$

C) $\frac{x^2}{b^2}+\frac{y^2}{b^2-a^2}=1$ D) none of these

45. The length of the latus rectum of the locus of the focus of the parabola for a = 4 and b = 3 is

A) $\frac{14}{3}$ B) $\frac{7}{2}$ C) $\frac{7}{3}$ D) $\frac{7}{\sqrt{2}}$

Paragraph for Question Nos. 46 and 47

Let C be a variable point in the first quadrant on the line $x + y = 1$ which intersects X and Y axes at A and B respectively. D and E are foot of perpendiculars from C on X and Y axes respectively. Let P be a point on the line segment joining D and E.

46. For all its possible positions, P will satisfy the inequality:

A) $xy \geq \frac{1}{16}$ B) $xy \leq \frac{1}{16}$ C) $xy \geq \frac{1}{32}$ D) $xy \leq \frac{1}{32}$

47. The area of the region traced by the point P, is

A) $\frac{1}{16}\ell n 2$ B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{1}{6}$

Paragraph for Question Nos. 48 and 49

Read the following write up carefully and answer the following questions:

A triangle ABC is such that a circle passing through vertex C, centroid G touches side AB at B. If AB = 6, BC = 4 then

48. The length of median through A is equal to

(A) $\frac{13}{2}$ (B) $\sqrt{42}$ (C) $2\sqrt{42}$ (D) none of these

49. Length of AC is equal to

(A) $\sqrt{17}$ (B) $2\sqrt{14}$ (C) $2\sqrt{17}$ (D) $\sqrt{14}$

Paragraph for Question Nos. 50 and 51

Read the following write up carefully and answer the following questions:

There is a point P inside an equilateral triangle ABC of side d whose distance from the vertices is 3, 4, 5. Rotate the triangle through 60^0 about C. Let A go to A' and P to P'.

50. The area of triangle PAP' is

(A) 8 (B) 12 (C) 6 (D) none of these

51. The value of d^2 is

(A) $25 + 12\sqrt{3}$ (B) $25 - 12\sqrt{3}$ (C) $5 + 12\sqrt{3}$ (D) $25 + 24\sqrt{3}$

Paragraph for Question Nos. 52 and 53

Read the following write up carefully and answer the following questions:

Let ABC be a triangle such that origin is its circumcentre & its base, BC, always passes through the point (4, 5). If the lateral sides (AB and AC) have x + y = 0 and x – 9y = 0 as their perpendicular bisectors, then the locus of A is a circle Γ, then,

52. Γ passes through which of the below points

A) (0, 0) B) (1, 41) C) $\left(\frac{4}{5}, \frac{3}{5}\right)$ D) (-1, -2)

53. The centre of Γ lies on the line

A) $x+5y=7$ B) $x-y=0$ C) $x=0$ D) $y=0$

MATRIX MATCHING TYPE (6 BITS)

54. Match the following List – I with List – II :

List – I		List – II	
(P)	Any chord of conic $x^2+y^2+xy=1$ passing through origin is bisected at point (p, q), then p + q is equal to	1.	1
(Q)	The base BC of ΔABC passes through the point P (1, 1) and its sides are bisected at right angles by x + y = 0 and x + 2y = 0. The locus of the vertex A is circle with radius equal to r, then $\frac{\sqrt{5}}{r}$ is	2.	5
(R)	The circle $x^2+y^2-4x-4y+4=0$ is inscribed in a triangle which has two of its sides along the coordinate axes. If the locus of circumcentre of triangle is $x+y-xy+\lambda\sqrt{x^2+y^2}=0$, then value of λ is equal to	3.	3
(S)	Let P be a point in xy plane, then minimum value of AP + BP + CP + DP is equal to ℓ, then $\frac{\ell}{3}$ is equal to (where A (0, 0), B (4, 3), C (3, 4) and D (–2, 11)	4.	0
		5.	7
		6.	$\sqrt{8}$

The correct option is:

(A) P → 4; Q → 6; R → 2; S → 1

(B) P → 1; Q → 4; R → 2; S → 3

(C) P → 4; Q → 3; R → 1; S → 2

(D) P → 6; Q → 5; R → 6; S → 6

55. Let C be curve represented by equation $5x^2 + 5y^2 - 8xy - 9 = 0$, match the following List-I with List-II

List-I		List-II	
(P)	Area enclosed by curve C is equal to …….. square unit	1.	1
(Q)	Let a tangent drawn at point P (other than vertex) on ellipse. If a line AP intersect the line passing through B perpendicular to above tangent at Q, then AQ is equal to (where A (–2, –2), B (2, 2))	2.	3π
(R)	If normal drawn at P(lying on C) cuts major axis at G and perpendicular from origin cuts normal at F, then PG·PF is equal to	3.	6
(S)	Let the line $y - x = \sqrt{2}$ touches C at P (α, β), then $8\left(\alpha^2 + \beta^2\right)$ is equal to	4.	5
		5.	4π
		6.	8

The correct option is:
(A) P → 4; Q → 6; R → 2; S → 1
(B) P → 2; Q → 1; R → 3; S → 6
(C) P → 4; Q → 6; R → 5; S → 2
(D) P → 2; Q → 3; R → 1; S → 6

56. Match the list :

List – I		List – II	
(P)	Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and is completely contained in it, is	(1)	16
(Q)	If the shortest distance between the curves $y^2 = 4x$ and $y^2 = 2x - 6$ is d then value of d^2 is	(2)	5
(R)	Let AB be a focal chord of $y^2 = 12x$ with focus S. The harmonic mean of lengths of segments AS and BS is	(3)	6
(S)	Tangents drawn from P meet the parabola $y^2 = 16x$ at A and B. If these two tangents are perpendicular then the least value of $\sqrt{AB}$ is,	(4)	4
		(5)	-1
		(6)	0

The correct option is:

(A) P → 4; Q → 2; R → 3; S → 5
(B) P → 4; Q → 2; R → 3; S → 4
(C) P → 4; Q → 2; R → 1; S → 6
(D) P → 4; Q → 3; R → 6; S → 5

57. A (–2, 0) and B (2, 0) are two fixed points and P is a point such that PA – PB = 2. Let S be the circle $x^2 + y^2 = r^2$. Then match the following.

	List – I		List – II
(P)	If r = 2, then the number of points P satisfying PA – PB = 2 and lying on $x^2 + y^2 = r^2$ is	(1)	2
(Q)	If r = 1, then the number of points satisfying PA – PB = 2 and lying on $x^2 + y^2 = r^2$ is	(2)	4
(R)	For r = 2 , the number of common tangents is	(3)	0
(S)	For $r = \frac{1}{2}$, the number of common tangents is	(4)	1
		(5)	5
		(6)	6

The correct option is

(A) P → 6; Q → 4; R → 3; S → 5
(B) P → 2; Q → 1; R → 1; S → 5
(C) P → 1; Q → 4; R → 3; S → 1
(D) P → 2; Q → 4; R → 3; S → 5

58. Consider an ellipse $(E)\ \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, centered at point 'O' and having AB and CD as its major and minor axes respectively. If S_1 be one of its foci, radius of in-circle of triangle OCS_1 be 1 unit and OS_1 = 6 units, then,

	List - I		List - II
P.	The length of the major axis of E is	**1.**	4
Q.	The maximum distance of normal from centre of E is	**2.**	$\frac{13}{2}$
R.	The distance between the directrix of E is	**3.**	$\frac{169}{12}$
S.	The length of the latus rectum of E is	**4.**	$\frac{25}{13}$

The correct option is:

(A) P → 2; Q → 1; R → 3; S → 4
(B) P → 1; Q → 2; R → 4; S → 3
(C) P → 2; Q → 1; R → 4; S → 3
(D) P → 1; Q → 3; R → 4; S → 2

59. Consider four conics defined as follows

$C_1 : \frac{x^2}{10} - \frac{y^2}{4} = 1;\ C_2 : \frac{x^2}{3} + \frac{y^2}{4} = 1;\ C_3 : \frac{y^2}{2} - x^2 = 1;\ \ C_4 : x^2 = 16y$

List – I		List – II	
P.	One of the tangent to C_1 is	1.	$y = 2x + 9$
Q.	One of the tangent to C_2 is	2.	$y = 2x + 7$
R.	One of the normal to C_3 is	3.	$y = 2x + 4$
S.	One of the normal to C_4 is	4.	$y = 2x + \frac{3}{\sqrt{5}}$
		5.	$y = 2x + 6$
		6.	$y = 2x + \frac{6}{\sqrt{7}}$

The correct option is:
(A) P → 5; Q → 3; R → 6; S → 1
(B) P → 2; Q → 4; R → 1; S → 3
(C) P → 5; Q → 3; R → 4; S → 1
(D) P → 5; Q → 1; R → 6; S → 3

THREE COLUMN MATRIX MATCHING

Answer questions 60, 61 and 62 by appropriately matching the information given in the three columns of the following table.

A tangent L is drawn at any point P in the first quadrant on the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$. It intersects the x - axis at A and y – axis at B, such that area of Δ OAB is minimum. A variable point $Q(\alpha,\beta)$ is taken on L from which pair of tangents QC and QD are drawn to the circle $x^2+y^2=12$ (O is origin)

Column - 1		**Column - 2**		**Column - 3**	
(I)	Coordinate of A : $\left(4\sqrt{2},0\right)$	**(i)**	Minimum area of $\Delta OAB=12$	**(P)**	Locus of circumcentre of ΔQCD is $3x+4y=6\sqrt{2}$
(II)	Coordinate of P : $P:\left(\frac{4}{\sqrt{2}},\frac{3}{\sqrt{2}}\right)$	**(ii)**	Number of points where L intersects circle $x^2+y^2=12$ is 2	**(Q)**	Equation of line $L:3x+4y-12\sqrt{2}=0$
(III)	Coordinate of B : $\left(0,4\sqrt{2}\right)$	**(iii)**	Number of points where L intersects circle $x^2+y^2=12$ is 0	**(R)**	Equation of locus of curve from where perpendicular tangents can be drawn to given ellipse is $x^2+y^2=24$
(IV)	Point of concurrency of variable chord of contact CD of the circle $x^2+y^2=12$ is $\left(\frac{3}{\sqrt{2}},\frac{4}{\sqrt{2}}\right)$	**(iv)**	Number of points on the curve $x^2+y^2=25$, from where perpendicular tangents can be drawn to $x^2+y^2=12$ is infinite	**(S)**	Line $x-y-\sqrt{7}=0$ intersecting the ellipse after reflection will pass through $\left(-\sqrt{7},0\right)$

60. Which of the following is **CORRECT** combination?
(A) (I) (i) (P) (B) (II) (iv) (S) (C) (IV) (iii) (P) (D) (III) (iii) (S)

61. Which of the following is **CORRECT** combination?
(A) (III) (i) (P) (B) (IV) (iv) (S) (C) (I) (iii) (R) (D) (II) (ii) (Q)

62. Which of the following is **INCORRECT** combination?
(A) (IV) (i) (P) (B) (IV) (iv) (R) (C) (II) (i) (P) (D) (I) (ii) (P)

Answer Q.63, Q.64 and Q.65 by appropriately matching the information given in the three columns of the following table.

Column-1: contains conics; Column-2: number of common tangents; Column-3: number of common Normal

Column – 1	Column – 2	Column - 3
(I) $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$	(i) 0	(P) 0
(II) $(y-1)^2 = \frac{4}{\sqrt{3}}\left(x + \frac{\sqrt{3}}{4}\right)$ and $\frac{\left(x+\frac{\sqrt{3}}{4}\right)^2}{9} + \frac{(y-1)^2}{3} = 1$	(ii) 1	(Q) 1
(III) $x^2 = y - 2$ and $y + x^2 - 2x + 3 = 0$	(iii) 2	(R) 2
(IV) $y^2 = -4x$ and $x^2 = -4y$	(iv) 3	(S) 3

63. Which of the following options is the only CORRECT combination?

A) (I) (iv) (Q) B) (I) (iv) (S) C) (II) (i) (S) D) (II) (ii) (S)

64. Which of the following options is the only CORRECT combination?

A) (IV) (i) (Q) B) (IV) (i) (P) C) (IV) (ii) (Q) D) (I) (iv) (P)

65. Which of the following options is the only INCORRECT combination?

A) (III) (iii) (Q) B) (II) (i) (S) C) (II) (iii) (S) D) (IV) (ii) (Q)

Answer Q.66, Q.67 and Q.68 by appropriately matching the information given in the three columns of the following table.

Column-1: locus of centre C of moving circle S

Column-2: locus of complex number Z

Column-3: required locus, match the following Column(s)

(**Note:** Here correct combination means if Column-1 is circle then Column-2 is also circle and Column-3 is also circle)

Column – 1	Column – 2	Column – 3
(I) If S touches always a line x = 4 and circle $(x-1)^2 + (y-2)^2 = 1$	(i) $\lvert z\rvert = \lvert z - 3 - 4i\rvert$	(P) ellipse
(II) If S touches both circles $(x-1)^2 + (y-2)^2 = 9$ & $(x-1)^2 + (y-1)^2 = 1$	(ii) $\lvert\lvert z\rvert - \lvert z - 4 - 5i\rvert\rvert = 3$	(Q) hyperbola
(III) If S touches both circles $x^2 + y^2 = 1$ and $(x-4)^2 + (y-5)^2 = 4$	(iii) $\lvert z - 1 - i\rvert + \lvert z - 1 - 2i\rvert = 4$	(R) parabola
(IV) If S touches both circles $x^2 + y^2 = 1$ and $(x-3)^2 + (y-4)^2 = 1$	(iv) $\lvert z - 1 - 2i\rvert = \lvert \mathrm{Re}(z-4) + 1\rvert$	(S) line

66. Which of the following is **CORRECT** combination?
(A) (I) (i) (P) (B) (I) (iv) (P) (C) (I) (iv) (R) (D) (I) (ii) (R)

67. Which of the following is **CORRECT** combination?
(A) (II) (ii) (R) (B) (II) (iii) (P) (C) (II) (iv) (P) (D) (II) (i) (R)

68. Which of the following is **CORRECT** combination?
(A) (III) (iii) (Q) (B) (III) (ii) (P) (C) (III) (ii) (Q) (D) (IV) (i) (R)

INTEGER TYPE QUESTIONS

69. Suppose that the circle $x^2+y^2-4ax-2ay+4a^2=0$ $(a>0)$ touches the lines $y=mx$ at A and B such that AB = 3. Then the value of $\frac{4a}{\sqrt{5}}$ is ---------------

70. Let $'\ell'$ be a tangent line at the point $T\left(t,\frac{t^2}{2}\right)(t>0)$ on the curve $C: y=\frac{x^2}{2}$. A circle S touches the curve C and the x – axis. Denote by $\left(x(t),y(t)\right)$ the center of circle S. If $\lim\limits_{r\to 0^+}\int_r^1 x(t)\,y(t)\,dt=\frac{\tan\frac{3\pi}{8}}{6a}$, then the value of a is ---------------

71. If ℓ_1 and ℓ_2 be the length of the focal chord other than latus rectum of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ which are perpendicular to each other, then $12\left(\frac{1}{\ell_1}+\frac{1}{\ell_2}\right)$ ------------------

72. If the normals to the curve $y=x^2$ at the points P, Q and R pass through the point $\left(0,\frac{3}{2}\right)$, then the radius of the circle circumscribing the triangle PQR is ---------------

73. The complete set of non-zero values of 'k' such that the equation $\left|x^2-7x+6\right|=kx$ is satisfied by at least one and at most three real value(s) of x is $\left(-\infty,\ \ell_1\right]\cup\left[\ell_2,\ \infty\right)$, then $\frac{\left|\ell_1-\ell_2\right|}{2}$ is equal to -----------

74. Two tangents to an ellipse $x^2+2y^2=2$ are drawn which intersect at right angles, let ℓ_1 and ℓ_2 be the intercepts on these tangents made by the circle $x^2+y^2=2$, then $\ell_1^2+\ell_2^2$ is equal to ----------

75. Consider the ellipse $4x^2+y^2-8x+4y+4=0$. Pair of tangents PT_1 and PT_2 are drawn to the ellipse from $P(-8,\ -2)$. Let S'_1 be the image of S_1 about PT_1 and S'_2 be the image of S_2 about PT_2, then if $\angle PS_1S'_2=\alpha$ and $\angle PS'_1S_2=\beta$, then $\frac{\alpha}{\beta}$ is equal to -----------

76. If a chord AB of the curve $y=\dfrac{3x-2}{x-2}$ is a normal to the curve at a point A, subtending an angle θ at the point $P(2,\ 3)$. Then the value of $\dfrac{\sin(\theta-\angle PAB)}{\sin(\theta+\angle PAB)}$ is equal to ------------

77. Let BD be the internal angle bisectors of angle B in triangle ABC with D on side AC. The circumcircle of triangle BDC meets AB at E, while the circumcircle of triangle ABD meets BC at F. if AE = 3, then CF is equal to ------------

78. Two circles C_1, C_2 of radii $\dfrac{1}{3}$ and $\dfrac{2}{7}$ touch each other externally and they both touch a unit circle C internally. A circle C_3 of radius R is inscribed to touch the circles C_1, C_2 externally and the circle C internally, then 19R is equal to --------------

79. Two circles of unequal radii have four common tangents. A transverse common tangent meets the direct common tangents at the points P & Q. If length of direct tangent (between the point of contacts) is 8 then length of PQ is

80. Let three lines $L_1, L_2 \,\&\, L_3$ belonging to the family $x-2y+6+\lambda(x-y+2)=0$, where λ is a parameter, be interior angle bisectors of ΔABC . If the equation $x+3y-4=0$ represents side AB of the triangle, then the value of $\left[\dfrac{\Delta}{\sum\left(r\cot\dfrac{A}{2}+a\right)}\right]$ is,

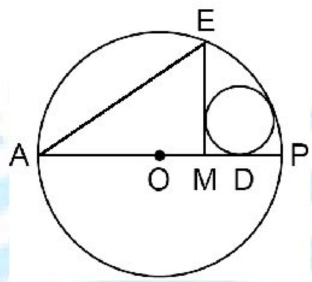
(Note: Symbols used have usual meaning in ΔABC and [.] denotes G.I.F.)

81. A ball moving around the circle $x^2+y^2-2x-4y-20=0$ in anticlockwise direction leaves it tangentially at P (–2, –2). After getting reflected from a straight line 'L', it passes through the centre of circle. If distance of P from line L is $\dfrac{5}{2}$ and 2α be the angle between incident ray and reflected ray, then the value of $10\cot 2\alpha\cos\alpha$ is equal to ------------

82. ΔABC is an equilateral triangle with circumcentre (1, 2) and vertex A lying on the line 5x + 2y = 4. A circle with centre I (2, 0) passes through the vertices B and C and intersects the sides AC and AB at D and E respectively. If area of quadrilateral BCDE is Δ, then find the value of $\dfrac{\Delta}{10\sqrt{3}}$ is,

83. All the three vertices of an equilateral triangle lie on the parabola $y = x^2$, and one of its sides has a slope of 2. The x-coordinates of the three vertices have a sum equal to $\frac{p}{q}$ where p and q are relatively prime positive integers, then the value of $(q - p)$ is........

84. Let 'O' be centre of circle as shown in figure and AE be a chord, diameter AP is drawn and M is foot of perpendicular drawn from E on AP. A circle is drawn such that it touches EM, AP at D and circle whose diameter is AP, then the value of $\frac{AE}{AD}$ is -----------



85. The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points A and B, there is a point P on this ellipse such that the area of ΔPAB is equal to $6\left(\sqrt{2} - 1\right)$, then the number of such points (P) is,

86. A straight line cuts the x-axis at point $A(1,0)$, and y-axis at point B, such that $\angle OAB = \alpha \left(\alpha > \frac{\pi}{4}\right)$. C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB', then the ratio of the areas of triangles ABB' and BB'C' is equal to ----------

87. If number of points common between $||x| - |y|| = 1$ and $2a\,|x|\,|y| + 1 = 2\,|x| + a|y|$ are 8, then the value of 3a (where $a < 1$) is ----------

NUMERICAL TYPE QUESTIONS

88. ABC is a right-angle triangle (right angled at B) inscribed in the parabola $y^2 = 4x$. The minimum length of the intercept cut off by the tangents at A and C to the parabola from y-axis is λ, then 1002.2λ is,

89. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Let perimeter of the rectangle ABCD is P, then 103.44P is ---------------

90. Let Δ ABC be equilateral on side AB produced, take a point P such that A lies between P and B. Consider 'a' as the length of sides of ΔABC, r_1 as the radius of incircle of Δ PAC, r_2 as the exradius of Δ PBC with respect to side BC, then $\frac{r_1 + r_2}{\sqrt{3}a}$ is equal to ----------

91. Let f: R $\to$R be a differentiable function such that f(0) = 0, f(1) = 1 and |f '(x)| < 2 $\forall$x $\in$ R, if a and b are real numbers such that the set of possible values of $\int_0^1 f(x)\,dx$ is the open interval (a, b), then (b – a) is -----------

92. The value of $\frac{k}{2}$ which minimizes $F(k)=\int_0^4 \left|x(4-x)-k\right|dx$ is ------------

93. Let ABC be right angle triangle, $\angle$B = 90°. The median through A and C are y = 3x + 1 and y = x + 1 respectively. If AC = 8, then area of triangle ABC is -------------

94. Coordinate of any point P is (4, 5). If A and B are variable points on straight lines y = x and y = 2x respectively. Then find minimum value of $(PA + PB + AB)^2$ is ________

95. Two circles are passing through point A of the ΔABC & one of the circle touches the side BC at B & other circle touches the side BC at C. If BC = 5 & $\angle$A = 30° then the product of radii of two circles is------------

96. Given circles C_1 & C_2 intersect at X & Y. Let I_1 be a line through the centre of C_1 & intersecting C_2 at points P & Q and let I_2 be the line through centre of C_2 intersecting C_1 at pts R & S . Let $C_1 : x^2+y^2-2x-2y+1=0$ & $C_2 : x^2+y^2-4x-4y+4=0$. If P, Q, R, S are concyclic with centre of circle passing through them being (α,β) , then $\tan^{-1}(\alpha+\beta)+\tan^{-1}(\alpha+\beta-1)$ is equal to $\tan^{-1}\left(\frac{10}{K}\right)$, then K is -------

97. A parallelogram is drawn to circumscribe the curve $x^2 + 4y^2 = 4$ such that two of its vertices are $\left(\sqrt{3},\sqrt{2}\right)$ and $\left(-\sqrt{3},-\sqrt{2}\right)$. If L is the sum of distances of the other two vertices from the centre of the curve then the value of $\frac{1}{L^2}$ is equal to

98. The least numbers A such that for any two squares of combined area 1 and a rectangle of area A exists such that the two squares can be packed in the rectangle (without interior overlap). You may assume that the sides of the squares are parallel to the sides of the rectangle, is -------------

99. P is a point on parabola $y^2 = 4x$ such that PS = 4 (S is the focus). Tangent is drawn at P to parabola which intersects tangent at vertex at T. A point R is taken on axis of parabola such that SR = 4 and R lies inside the parabola. The area of quadrilateral PRST is q, then the value of $\sqrt{3}q$ is,

100. There is a point $A(x_1, y_1)$ on the curve $f(x)=x^2$ and a point $B(x_2, y_2)$ on the curve $g(x)=-\frac{8}{x}$ where $x_1, x_2 > 0$. If the line AB is tangent to both these curve at these points respectively, then the value of $x_1+y_1+x_2$ is………

KEY

1	2	3	4	5	6	7	8	9	10
C	**B**	**A**	**A**	**A**	**A**	**A**	**C**	**A**	**C**
11	12	13	14	15	16	17	18	19	20
C	**D**	**D**	**D**	**C**	**B**	**AC**	**BC**	**AC**	**ABC**
21	22	23	24	25	26	27	28	29	30
AB	**A**	**AC**	**AC**	**ABC**	**AD**	**AB**	**D**	**ABD**	**ABC**
31	32	33	34	35	36	37	38	39	40
ABCD	**AC**	**BC**	**D**	**D**	**B**	**C**	**B**	**C**	**A**
41	42	43	44	45	46	47	48	49	50
A	**A**	**B**	**C**	**A**	**B**	**D**	**B**	**B**	**C**
51	52	53	54	55	56	57	58	59	60
A	**A**	**C**	**C**	**D**	**B**	**C**	**A**	**A**	**A**
61	62	63	64	65	66	67	68	69	70
D	**B**	**A**	**C**	**B**	**C**	**B**	**C**	**3**	**5**
71	72	73	74	75	76	77	78	79	80
7	**1**	**7**	**4**	**1**	**3**	**3**	**2**	**8**	**1**
81	82	83	84	85	86	87	88	89	90
5	**9**	**8**	**1**	**3**	**2**	**2**	**4008.80**	**8275.20**	**00000.50**
91	92	93	94	95	96	97	98	99	100
0.75	**1.50**	**00010.66**	**00016.40**	**00025.00**	**00001.25**	**0.05**	**1.20**	**18**	**21**

HINTS

1. Any point on the line through P (a, 2) is $\left(a+r\cos\theta, 2+r\sin\theta\right)$ and meets the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ at two distinct point A and D

$$\Rightarrow \frac{\left(a+r\cos\theta\right)^2}{9}+\frac{\left(2+r\sin\theta\right)^2}{4}=1$$

$$\Rightarrow \left(4\cos^2\theta+9\sin^2\theta\right)r^2+4\left(2a\cos\theta+9\sin\theta\right)r+4a^2=0$$

$$\Rightarrow PA.PD=\frac{4a^2}{\left(4\cos^2\theta+9\sin^2\theta\right)}$$

also line meets the axes of B and C $\Rightarrow PB.PC=\frac{2a}{\sin\theta.\cos\theta}$

since PA, PB, PC and PD are in GP.

$$\Rightarrow \frac{4a^2}{4\cos^2\theta+9\sin^2\theta}=\frac{2a}{\sin\theta.\cos\theta}$$

$$\Rightarrow 2a\sin 2\theta+5\cos 2\theta=13$$

$$\Rightarrow -1\le\frac{13}{\sqrt{4a^2+25}}\le 1\Rightarrow a\le -6 \text{ or } a\ge 6$$

2. $\because 2\cos\theta_1 = \frac{8}{5} \Rightarrow \cos\theta_1 = \frac{4}{5} \Rightarrow \tan\frac{\theta_1}{2} = \sqrt{\frac{1-\frac{4}{5}}{1+\frac{4}{5}}} = \frac{1}{3}$ and $e = \frac{1}{2}$

$\therefore \tan\frac{\theta_2}{2} \times \frac{1}{3} = \frac{\frac{-1}{2}}{\frac{3}{2}}$

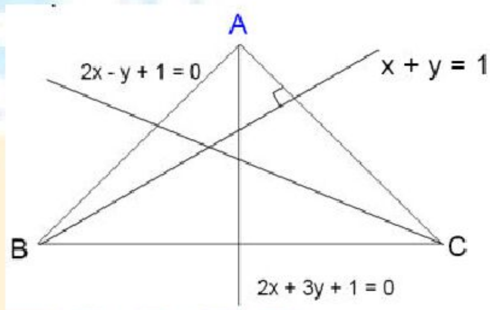
$\frac{\theta_2}{2} = \frac{3\pi}{4} \Rightarrow \theta_2 = \frac{3\pi}{2}$

$\therefore B \equiv \left(0, -\sqrt{3}\right) \Rightarrow AB = 2 + 2 - \frac{1}{2} \times \frac{8}{5} = 4 - \frac{4}{5} = \frac{16}{5}$

3. Slope of AC $= -1$

Let slope of BC = m

Then $\left|\frac{2-1}{1+2}\right| = \left|\frac{m-2}{1+2m}\right|, \Rightarrow m = 7$



Let equation of BC is y = 7x + C

So, $B = \left(\frac{1-C}{8}, \frac{7+C}{8}\right), C \equiv \left(\frac{1-C}{5,}, \frac{7-2C}{5}\right)$

Mid-point of BC lie on 2x + 3y + 1= 0

Hence, $C = \frac{379}{59}$

4. $x + \sqrt{x^2+1} = \frac{1}{y+\sqrt{y^2+1}}$

$\ell n\left(x+\sqrt{x^2+1}\right) = \ell n\left(\frac{1}{y+\sqrt{y^2+1}}\right)$

$x + y = \sqrt{x^2+1} + \sqrt{y^2+1} \;.....(1)$

Similarly, $y + \sqrt{y^2+1} = \frac{1}{x+\sqrt{x^2+1}}$

$x + y = -\sqrt{x^2+1} - \sqrt{y^2+1} .....(2)$

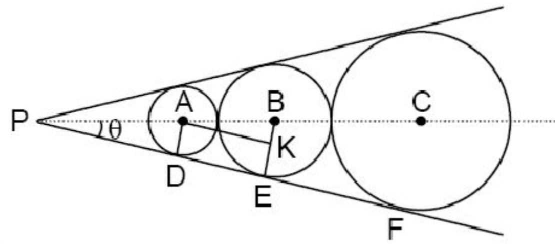
Adding 1 and 2, we get 2x + 2y = 0

Image of xy=1 is xy = 1 itself.

5. Since the parabolas are the same shape, the "rolling" parabola will always be the reflection of the stationary parabola over the tangent line at the point of contact. Hence, the focus of rolling Parabola will be reflection of Focus of fixed Parabola about any arbitrary common tangent to both these Parabola. As reflection of focus about any tangent to Parabola lies on its Directrix and hence Focus of rolling Parabola lies in directrix of fixed Parabola. Hence, Focus of rolling Parabola must lie on

directrix of $y = x^2 - x + 1$. Hence, locus is $y = 1$

6. $DE = AK = \sqrt{(r_2 + r_1)^2 - (r_2 - r_1)^2} = 2\sqrt{r_1 r_2}$



Similarly, $EF = 2\sqrt{r_2 r_3}$

$\angle APD = \theta$ and $PD = x$ then

$\tan\theta = \frac{AD}{PD} = \frac{BE}{PE} = \frac{CF}{PF} = \frac{r_1}{x}$

$= \frac{r_2}{x + 2\sqrt{r_1 r_2}} = \frac{r_3}{x + 2\sqrt{2r_1 r_2} + 2\sqrt{r_2 r_3}}$

Hence, $\frac{r_2 - r_1}{2\sqrt{r_1 r_2}} = \frac{r_3 - r_2}{2\sqrt{r_2 r_3}} \Rightarrow r_2 = \sqrt{r_1 r_3} \Rightarrow r_2 = \sqrt{2 \times 8} = 4$

7. $\frac{R'Q}{CA} = \frac{PQ}{AB} \Rightarrow PQ = \frac{102 - d}{102}.85$

$\frac{R''R}{BC} = \frac{PR}{AB} \Rightarrow PR = \frac{90 - d}{90}.85$

$QR = d = \frac{102 - d}{102}.85 + \frac{90 - d}{90}.85$

$\Rightarrow d\left(1 + \frac{85}{102} + \frac{85}{70}\right) = 85 + 85 \Rightarrow d\left(1 + \frac{5}{6} + \frac{17}{18}\right) = 170 \Rightarrow d = \frac{170.18}{50} \Rightarrow \frac{5d}{34} = 9$

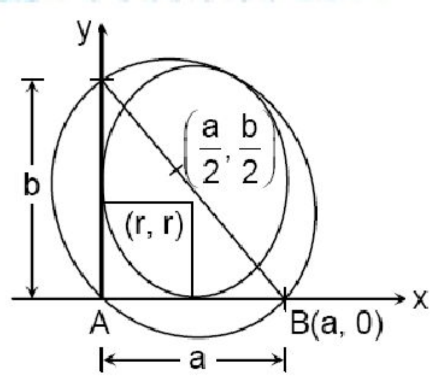
8. If any tangent to an ellipse is cut by the tangents at the ends of major axis in the points T and T' then the circle, whose diameter is TT', will pass through the foci of the ellipse.

Hence, solving $y = x$ with circle, we get

$S_1 \equiv \left(2\sqrt{2}, 2\sqrt{2}\right)$ *and* $S_2 \equiv \left(-2\sqrt{2}, -2\sqrt{2}\right)$

$\Rightarrow S_1 S_2 = 8$

9. Equation of $S = \left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2 + b^2}{4}$



$x^2 + y^2 - ax - by = 0$

And circle with center (r, r) and radius 'r' touches AB and AC.

Its equation is,

$(x - r)^2 + (y - r)^2 = r^2$

$x^2 + y^2 - 2xr - 2yr + r^2 = 0$

Both circle touches internally or externally

Using $\sqrt{\left(r-\frac{a}{2}\right)^2+\left(r-\frac{b}{2}\right)^2}=\sqrt{\left(\frac{a^2}{4}+\frac{b^2}{4}\right)}\pm r$

we get, area of $\Delta ABC=\frac{r_1r_2}{4}$

10. Equation of tangent at, (i, yj)

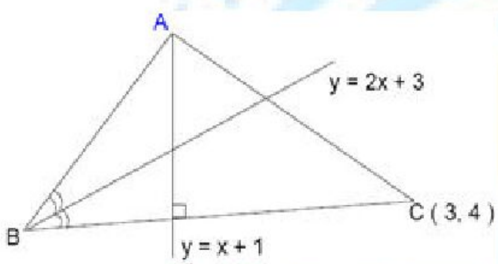
$y=(2i+b)x-i^2+c$

Also, at $(i+1, y_{i+1}), y=(2i+1)+b)x-(i+1)^2+c$

Point of intersection is $x=\frac{2i+1}{2}\Rightarrow i=\frac{2x-1}{2}$

Put this 'i' in any tangent, we get $y=x^2+bx+c-\frac{1}{4}$

11. Equation of BC is $x+y=7$ so $B=\left(\frac{4}{3},\frac{17}{3}\right)$



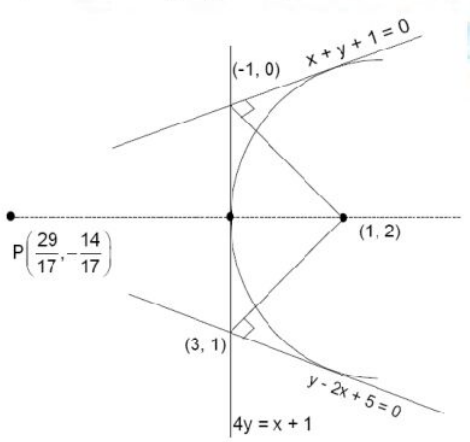
Let slope of AB is m so, $\left|\frac{m-2}{1+2m}\right|=\left|\frac{2+1}{1-2}\right|, \Rightarrow m=-\frac{1}{7}$

So equation of AB is 3x + 21y – 123 =0

So, $A\equiv\left(\frac{17}{4},\frac{21}{4}\right)$ mid-point of $AB\equiv\left(\frac{67}{24},\frac{131}{24}\right)$

Equation of median through C is 7x + y = 25

12. Normal through given point be the axis of parabola, $y-2=-4(x-1)$ or $4x+y=6$



13. $(x-2y-4)^2=12(2x+y)$

14. Image of (2, 1) in $2x - y + 1 = 0$ is

$$\frac{x-2}{2}=\frac{y-1}{-1}=\frac{-2(4-1+1)}{4+1}=\frac{-8}{5}$$

$$\Rightarrow x=2-\frac{16}{5}=\frac{-6}{5}$$

$$y=\frac{8}{5}+1=\frac{13}{5}$$

$$\therefore \text{Slope of directrix } =\frac{\frac{13}{5}-3}{\frac{-6}{5}-1}=\frac{-\frac{2}{5}}{-\frac{11}{5}}=\frac{2}{11}$$

$$\therefore \text{ Equation of axis, } y-1=\frac{-11}{2}(x-2)$$

$$\Rightarrow 2y-2=-11x+22$$

$$\Rightarrow 11x+2y=24$$

15. Let foot of perpendicular from (3, -1) on the directrix be (x_1, y_1)

So, $\frac{x_1-3}{2}=\frac{y_1+1}{1}=\frac{-(6-1-3)}{5}$

$$\frac{x_1-3}{2}=\frac{y_1+1}{1}=\frac{-2}{5}$$

$$x_1=3-\frac{4}{5};y_1=-1-\frac{2}{5}$$

$$x_1=\frac{11}{5};y_1=-\frac{7}{5}$$

Now image of (x_1, y_1) about tangent is focus, which is (x_2, y_2)

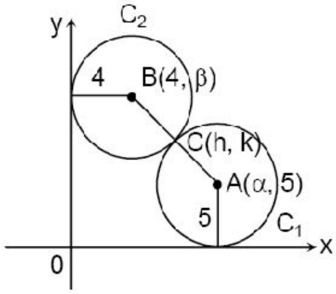
$$\frac{x_2-\frac{11}{5}}{1}=\frac{y_2+\frac{7}{5}}{1}=\frac{2\left(\frac{11}{5}-\frac{7}{5}-2\right)}{2}=\frac{6}{5}$$

$$x_2=\frac{17}{5};y_2=\frac{6}{5}-\frac{7}{5}=-\frac{1}{5}$$

Focus is $\left(\frac{17}{5},-\frac{1}{5}\right)$

16. C (h, k) divides AB into the ratio 5:4

$$\therefore (h,k)\equiv\left(\frac{4\alpha+20}{9},\frac{5\beta+20}{9}\right)$$



$AB = 9 \Rightarrow (\alpha-4)^2 + (\beta-5)^2 = 9^2$

$\Rightarrow \frac{(h-4)^2}{4^2} + \frac{(k-5)^2}{5^2} = 1$

Locus of point of contact is $\frac{(x-4)^2}{16} + \frac{(y-5)^2}{25} = 1$

We shift the axes to (4, 5), so that curve become $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and point (9, 5) become (5, 0)

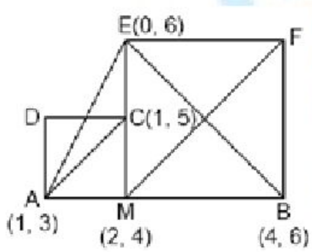
Chord of contact is $x = \frac{16}{5}$

Solving with conic, we get Q and R as $\left(\frac{16}{5}, \pm 3\right)$

Area of $\Delta PQR = \frac{27}{5}$

17. Conceptual

18. As shown in the figure since $C(1, 5)$ is the orthocenter of triangle ΔAEB, similarly for the other side, the coordinate is $C(3, 3)$



19. Equation of normal to hyperbola $xy = c^2$ at $\left(ct, \frac{c}{t}\right)$ which passes through (h, k) is

$ct^4 - ht^3 + kt - c = 0$

Root of this equation is t_1, t_2, t_3 and t_4

Let equation of circle $x^2 + y^2 - 2gx - 2fy + p = 0$

If $\left(ct, \frac{c}{t}\right)$ lies to this, then $c^2t^4 - 2gct^3 + pt^2 - 2fct + c^2 = 0$

Its roots are $t_1 t_2 t_3$ and t_4'

We get $t_4 = -t_4'$ also $\frac{c}{t_4} - \frac{c}{t_4'} = k - 2f$

So, locus of center is $4c^2 = (h-2g)(k-2f)$

centre lies on hyperbola $\left(x-\frac{h}{2}\right)\left(y-\frac{k}{2}\right) = c^2$

$C\left(\frac{h}{2}, \frac{k}{2}\right)$ and $c = \sqrt{2}$

20. $\sqrt{(\alpha+1)^2 + (\alpha+2-2)^2} + \sqrt{(\alpha-2)^2 + (\alpha+2-5)^2}$

Use concept of Optimization for PA+PB to be minimum using line equation as $y = x+2$ and $P(\alpha, \alpha+2)$ is any point on this line and $A \equiv (2, 5) \& B \equiv (-1, 2)$

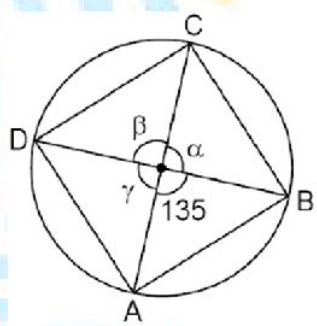
21. Foot of perpendicular from $P(4, -3)$ to the directrix is $(3, -1)$

Now, find its image about the tangent $\frac{\alpha-3}{1} = \frac{\beta+1}{-1} = \frac{-2(3+1-7)}{2}$

$\alpha = 6,\ \beta = -4$. So, focus $\equiv (6, -4)$

Now, distance of focus from the directrix = Length of Semi Latus Rectum $= \frac{|6+8-5|}{\sqrt{5}} = \frac{9}{\sqrt{5}}$

22. $\because\ \alpha, \beta, \gamma < \pi$ and $y = \sin x$ is concave downward in $(0, \pi)$



So, $\frac{\sin\alpha + \sin\beta + \sin\gamma}{3} \le \sin\left(\frac{\alpha+\beta+\gamma}{3}\right)$

$\Rightarrow \frac{\sin\alpha + \sin\beta + \sin\gamma}{3} \le \sin\left(75^0\right)$

So, area of ABCD $\le \frac{1}{2}\sin\left(135^0\right) + \frac{1}{2}(\sin\alpha + \sin\beta + \sin\gamma)$

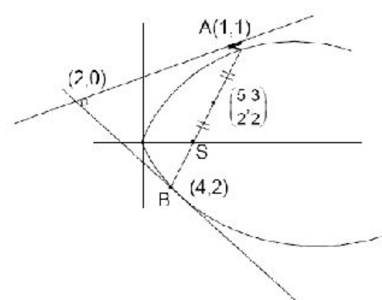
$= \frac{5+3\sqrt{3}}{4\sqrt{2}}$

Equality holds if $\alpha = \beta = \gamma = 75^0$

23. $\because$ Tangents are perpendicular so, they intersect on directrix.

Point of intersection on directrix $\equiv (2, 0)$ and mid-point of $(1, 1)$ and $(4, 2)$ is $\left(\frac{5}{2}, \frac{3}{2}\right)$

$\therefore$ Slope of axis $= \frac{\frac{3}{2}-0}{\frac{5}{2}-2} = 3$



Equation of directrix, $y = -\frac{1}{3}(x-2)$

$x + 3y = 2$

AB is focal chord.

BS = ($\perp$, distance from B on directrix) $= \dfrac{4+6-2}{\sqrt{10}} = \dfrac{8}{\sqrt{10}}$

AS = ($\perp$, distance from A on directrix) $= \dfrac{1+3-2}{\sqrt{10}} = \dfrac{2}{\sqrt{10}}$

So, focus divides AB in 1 : 4 ratios. So $S \equiv \left(\dfrac{8}{5}, \dfrac{6}{5}\right)$

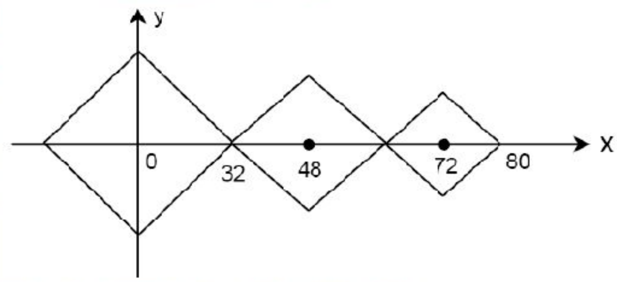
24. $a_2 = a_1 + \dfrac{3}{2}b_1 = 48$, $b_2 = \dfrac{b_1}{2} = 16$

$a_3 = 48 + \dfrac{3}{2} \times 16 = 72$, $b_3 = \dfrac{16}{2} = 8$

So, the three loops from $i = 1$ to $i = 3$ are alike Now, area of i^{th} loop (square) $= \dfrac{1}{2}(\text{diagonal})^2$

$A_i = \dfrac{1}{2}(2b_i)^2 = 2b_i^2$

$\therefore \dfrac{A_{i+1}}{A_i} = \dfrac{2(b_{i+1})^2}{2(b_i)^2} = \dfrac{1}{4}$



So, the areas form a G.P. series

$\therefore$ sum of the G.P. up to infinite term is

$\dfrac{A_i}{1-r} = \dfrac{2(32)^2}{1-\dfrac{1}{4}} = \dfrac{8}{3}(32)^2$ sq.units

25. For area of parallelogram PQRS to be least line L makes angle $\dfrac{\pi}{4}$ with L_1 and L_2

26. $y^2 = 16x \Rightarrow a = 4$

Slope of tangents at t is $\dfrac{1}{t}$

$\tan\theta_1 = \dfrac{1}{t_1}$, $\tan\theta_2 = \dfrac{1}{t_2}$, $\tan\theta_3 = \dfrac{1}{t_3}$

t_1, t_2, t_3 are in A.P. with $d = 3$

$t_2 - t_1 = 3$

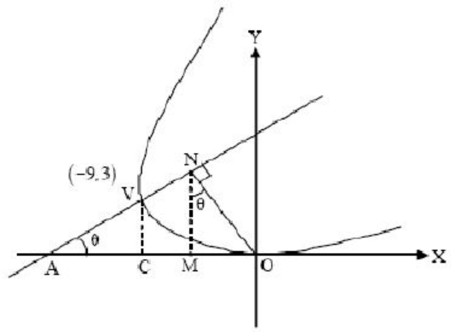
$t_3 - t_2 = 3$

Area of $\Delta PQR = \dfrac{1}{2}\begin{vmatrix} at_1t_2 & a(t_1+t_2) & 1 \\ at_2t_3 & a(t_2+t_3) & 1 \\ at_3t_1 & a(t_3+t_1) & 1 \end{vmatrix}$

$= a^2d^3 = 16 \times 27 = 432$

Area of ΔABC is $= 2.\Delta PQR = 864$.

27.



VC = 3 units, CO = 9 units

As per property : $AV = VN$ and $\frac{AN}{AV} = 2 = \frac{MN}{VC}$

$\Rightarrow OM = 6\tan\theta,\ AC = CM = 3\cot\theta \Rightarrow 9 = 3\left(2\tan\theta + \frac{1}{\tan\theta}\right)$

$\Rightarrow m = 1$ or $m = \frac{1}{2}$

$\Rightarrow$ Equation of Axis can be

$(y-3) = 1(x+9)$ or $(y-3) = \frac{1}{2}(x+9)$

28. $OQ^2 = OP^2 + PQ^2$

$\Rightarrow t_1^2 + 4t_1^2 = 5$

$\Rightarrow t_1 = 1 (\because t_1 > 0)$

$\Rightarrow Q \equiv (1, 2)$

$PQ = 1 \Rightarrow \cos\theta + 2\sin\theta = 2$

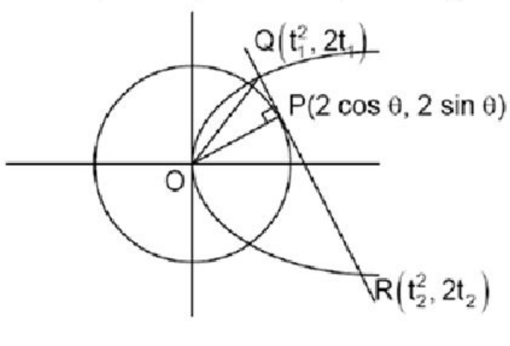
$\Rightarrow \cos\theta = \frac{4}{5}, \sin\theta = \frac{3}{5}$

$\Rightarrow P \equiv \left(\frac{8}{5}, \frac{6}{5}\right)$

Now R lies on tangent $4x + 3y = 10$

$\Rightarrow t_2 = \frac{-5}{2} (\because t_2 < 0)$

$\Rightarrow R \equiv \left(\frac{25}{4}, -5\right)$ and $S \equiv \left(-\frac{5}{2}, -\frac{3}{2}\right)$



29. $N(-2,2)$

$\angle NOM = 45^0$

$ON = AN = 2\sqrt{2}$

$OA = 4$

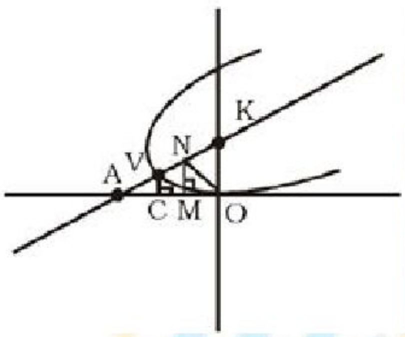
$NK = 2\sqrt{2} = 2a$ (semi L.R.)

$\Rightarrow \ a = \sqrt{2}$

eqn. of NK : $y = x + 4$

$\therefore \ A:(-4,\ 0)$

$V:(-3,1) \qquad \{\because (AV = VN)\}$



30. Conceptual

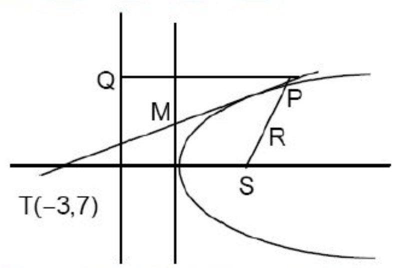
31. Equation of tangent at P is $x + y = 4$

Clearly mirror image of $R(-5,\ 5)$ lies on line PQ.

Now mirror image R' of R is,

$\Rightarrow \dfrac{\alpha'+5}{1} = \dfrac{\beta'-5}{1} = \dfrac{-2(-5+5-4)}{2} = 4$

$\Rightarrow (\alpha',\ \beta') \equiv (-1,\ 9)$



Let PM cuts the axis at T; As M is midpoint PT

$\Rightarrow T$ is $(-3,\ 7)$

We know that $SP = ST$ and $\angle SMP = \dfrac{\pi}{2}$

Equation $SP \equiv y - 3 = -\dfrac{1}{3}(x-1)$

$\Rightarrow x + 3y - 10 = 0$

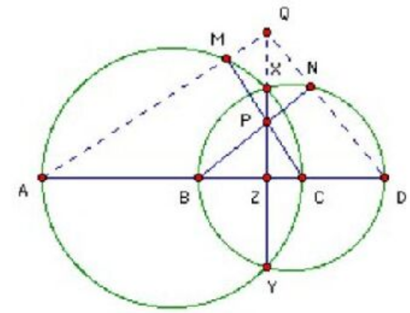
Let $S \equiv (10 - 3\beta,\ \beta)$

Again $TS \parallel PQ \Rightarrow \dfrac{\beta - 7}{13 - 3\beta} = \dfrac{9-3}{-1-1} = -3$

$\therefore$ Focus is $(-2,\ 4)$

32. $B(4,\ 0)$ is focus and $x = 0$ is directrix of parabola

33. Let DN meet XY at Q.



Angle QDZ = 90º - angle NBD = angle BPZ.
So triangles QDZ and BPZ are similar.
Hence QZ/DZ = BZ/PZ, or QZ = BZ·DZ/PZ.
Let AM meet XY at Q'. Then the same argument shows that Q'Z = AZ·CZ/PZ.
But BZ·DZ = XZ·YZ = AZ·CZ,
so QZ = Q'Z. Hence Q and Q' coincide.

Since, XY, AM and ND are concurrent, $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$a\left(bc - a^2\right) - b\left(b^2 - ac\right) + c\left(ab - c^2\right) = 0$$

$$abc - a^3 - b^3 + abc + abc - c^3$$

$$3abc - a^3 - b^3 - c^3 = 0$$

$$a^3 + b^3 + c^3 - 3abc = \left(a + b + c\right)\left(a + b\omega + c\omega^2\right)\left(a + b\omega^2 + c\omega\right) = 0$$

Or, $a^3 + b^3 + c^3 - 3abc = \left(c + a + b\right)\left(c + a\omega + b\omega^2\right)\left(c + a\omega^2 + b\omega\right) = 0$

34-35. Let $P(h, k)$ be a point with the given property and assume h, $k > 0$

Now, without loss of generality, we take $PC > PA$

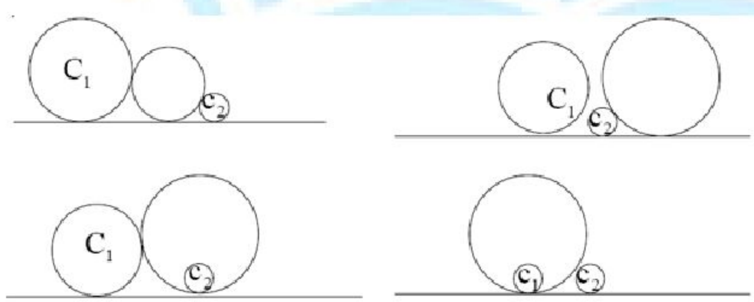
So, $\sqrt{2}PC = PB + PD$

$$\Rightarrow 2h^2 + 2(k + a)^2 = (h - a)^2 + (h + a)^2 + 2k^2 + 2\sqrt{(h + a)^2 + k^2} \times \sqrt{(h - a)^2 + k^2}$$

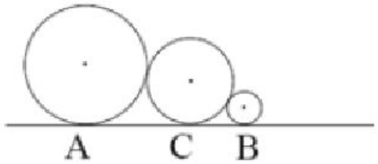
$\Rightarrow$ locus comes out to be $x^2 + y^2 = a^2$

So, circumcircle of square ABCD will be locus of point P.

36.

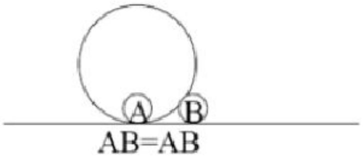


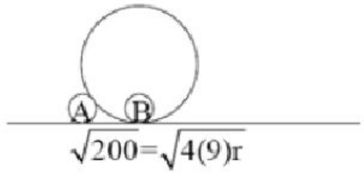
37.



$AB = AC + BC$

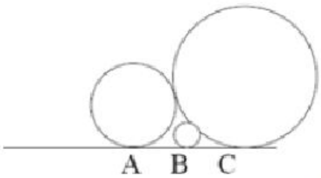
$$= \sqrt{225 - 25} = \sqrt{4r(9)} + \sqrt{4r(4)}$$





$\sqrt{4r(4)} = \sqrt{200}$, $r = \frac{50}{9}$

$r = \frac{25}{2}$



$AC - BC = AB$

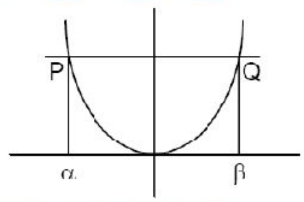
$\sqrt{4(9)r} - \sqrt{4(4)r} = \sqrt{200}$

$2\sqrt{r} = \sqrt{200}$

$r = 50$

38-39. Any two point on $y = x^2$ is $P(\alpha, \alpha^2)$; $Q(\beta, \beta^2)$

Equation of PQ, $y - \alpha^2 = (\alpha + \beta)(x - \alpha)$



$y = (\alpha + \beta)x - \alpha\beta$

Required area $\int_{\alpha}^{\beta}\left((\alpha + \beta)x - \alpha\beta - x^2\right)dx$

$\Rightarrow \beta - \alpha = 2$

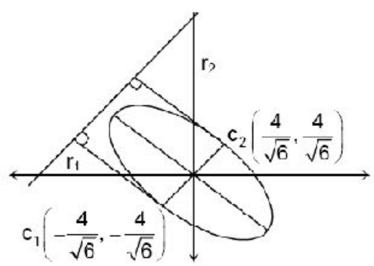
Pair of tangents from origin are $y = 2x$ and $y = -2x$

Area $\int_{0}^{1}\left(\left(x^2 + 1\right) - 2x\right)dx = \frac{2}{3}$

40-41. For the ellipse $x^2 + y^2 + xy = 8$, $C \equiv (0, 0)$

For the Parabola $\frac{8}{\sqrt{m^2 + 1}} = \frac{8}{\sqrt{3}}$: $(\because C_1C_2 = r_1 + r_2)$

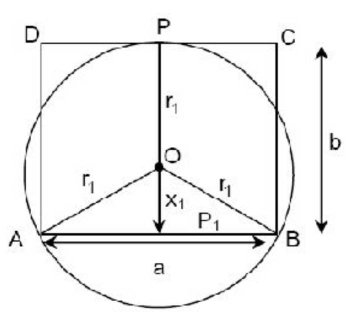
For two Parabola (s) : $C_1C_2 < r_1 + r_2$



42-43. $r_1 = b - x_1 = OP = OA$

$AP_1 = \frac{a}{2}$

$r_1^2 = x_1^2 + \left(\frac{a}{2}\right)^2 = (b - x_1)^2$



$$x_1^2+\frac{a^2}{4}=b^2+x_1^2-2bx_1$$

$$x_1=\frac{4b^2-a^2}{8b}$$

$$r_1=b-x_1=\frac{4b^2+a^2}{8b}$$

Similarly, $r_2=\dfrac{4a^2+b^2}{8a}$

$$r_1+r_2=\frac{a^3+b^3+4ab(a+b)}{8ab}$$

$$=\frac{(a+b)\left(a^2+3ab+b^2\right)}{8ab}$$

$$=\left(\frac{a+b}{8}\right)\frac{\left[(a-b)^2+5ab\right]}{ab}$$

But $(a-b)^2\ge 0$

$$r_1+r_2\ge\frac{(a+b)}{8}.\frac{5ab}{ab}$$

$$\Rightarrow r_1+r_2\ge\frac{5(a+b)}{8}$$

44. Let $A\equiv(-a,0)$, $B\equiv(a,0)$, the equation of the circles are $x^2+y^2=a^2$ and $x^2+y^2=b^2$ ….(1)

Any tangent to (1) is $y=mx+b\sqrt{1+m^2}$

Let (h,k) be the coordinates of the focus, A and B lie on the parabola.

$$\therefore\sqrt{1+m^2}\sqrt{(h+a)^2+k^2}=-ma+b\sqrt{1+m^2}$$

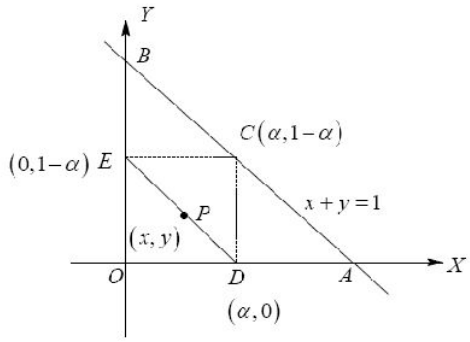
and $\sqrt{1+m^2}\sqrt{(h-a)^2+k^2}=ma+b\sqrt{1+m^2}$

Eliminating m from the above equations, we get the locus as $\dfrac{x^2}{b^2}+\dfrac{y^2}{b^2-a^2}=1$

45. For $a=4, b=3$, the curve becomes $\dfrac{x^2}{9}-\dfrac{y^2}{7}=1$

The length of $LR=\dfrac{2B^2}{A}=\dfrac{2\times 7}{3}=\dfrac{14}{3}$

46-47.



$$\frac{x}{\alpha}+\frac{y}{1-\alpha}=1$$

$$AM \geq GM$$

$$\frac{\frac{x}{\alpha}+\frac{y}{1-\alpha}}{2} \geq \sqrt{\frac{xy}{\alpha(1-\alpha)}}$$

$$\Rightarrow xy \leq \frac{\alpha(1-\alpha)}{4}$$

$$xy \leq \frac{1}{16}$$

Region traced by point P is $(y-x-1)^2 \geq 4x$

$\therefore$ Area $= \int_0^1 \left(x+1-2\sqrt{x}\right) = \frac{1}{6}$

48-49. Let median through A meet BC and D and circle at F

Let $GD = x$, $DF = y$ then $AG.AF = AB^2$

$2x(3x+y) = 36$ ...(1)

$xy = 4$

$3x^2 + 4 = 18$

$x^2 = \frac{14}{3}$

So, $AD = 3x = \sqrt{42}$

Also, $AC^2 + AB^2 = 2\left(AD^2 + BD^2\right)$

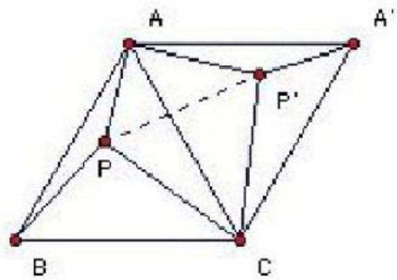
$AC^2 + 36 = 2(42+4) = 2 \times 46 = 92$

$AC^2 = 56$

$AC = 2\sqrt{14}$

50-51. Take the triangle to be ABC and the point P. Let PA = 3, PB = 4, PC = 5. Rotate the triangle and P through 60^0 about C. Let A go to A' and P to P'. Then $CP = CP'$ and $\angle PCP' = 60^0$, so PCP' is equilateral with side 5. So PAP' is a 3, 4, 5 triangle and hence $\angle PAP' = 90^0$. Hence $\angle PAB + \angle P'AA' = 120^0 - 90^0 = 30^0$. Hence $\angle APB = 150^0$. Now applying the cosine formula to

PAB we have $d^2 = 3^2 + 4^2 - 24\cos 150^0 = 25 + 12\sqrt{3}$.



52-53. Let $A(h, k)$, $C(x_0, y_0)$, $B(x_1, y_1)$

$h + x_0 = 9(k + y_0)$ and $h^2 + k^2 = x_0^2 + y_0^2 \Rightarrow x_0 = \frac{1}{41}(40h + 9k)$, $y_0 = \frac{1}{41}(9h - 40k)$

$h + x_1 = -(k + y_1)$ and $h^2 + k^2 = x_1^2 + y_1^2 \Rightarrow x_1 = -k,\ y_1 = -h$

$\frac{5 - y_1}{4 - x_1} = \frac{5 - y_0}{4 - x_0} \Rightarrow 4h^2 + 4k^2 + 41k = 0$

54. Equation of circle is $3x^2 + 3y^2 + 2x + 4y = 0$ and minimum value of $PA + BP + CP + DP = 15$

55. $\frac{(x+y)^2}{18} + \frac{(x-y)^2}{2} = 1$

56. (A) $(x-h)^2 + y^2 = (h-1)^2$

$y^2 = 4x$

$(x-h)^2 + 4x = (h-1)^2$

Apply $\Delta = 0$

(B) Equation of common normal to $y^2 = 4x, y^2 = 2x - 6$ is $y = m(x-3) - m - \frac{1}{2}m^3$

Equate $c = -2am - am^3$ for both the curves

$-4m - \frac{1}{2}m^3 = -2m - m^3$

$\frac{m^3}{2} - 2m = 0 \Rightarrow m = 0, m = \pm 2$

(C) Length of the semi latus rectum is the H.M between segments of the focal chord.

(D) Tangents at the ends of focal chord are perpendicular.

57. Basic definition of hyperbola and common tangent with circle.

58. Area $\Delta OCS_1 = \frac{1}{2} \times b \times 6 = 3b$

Semi – perimeter $= \frac{a+b+6}{2}$

Inradius $= \frac{\Delta}{S} = 1 \Rightarrow 5b = 6 + a$

Also, $b^2 + 36 = a^2$

Solving, we get $a=\frac{13}{2}, b=\frac{5}{2}$

Maximum distance of normal from center $=|a-b|=\frac{13-5}{2}=4$

Length of latus rectum $=\frac{2b^2}{a}=\frac{2\times\frac{25}{4}}{\frac{13}{2}}=\frac{25}{13}$

$e=\sqrt{\frac{a^2-b^2}{a^2}}=\frac{12}{13}$

Distance between directrix $=\frac{2a}{e}=\frac{2\times\frac{13}{2}}{\frac{12}{13}}=\frac{169}{12}$

59. Use Slope Form.

60-62. Equation of tangent is $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$

$a=4, b=3$

$\Rightarrow A:(4\sec\theta,0),\ B:(0,3\cos ec\theta)$

$\Delta OAB=\frac{1}{2}(45\sec\theta)(3\cos ec\theta)=\frac{12}{\sin 20}$

$(\Delta OAB)_{\min}=12 \text{ where } \theta=\frac{\pi}{4}$

$P:\left(\frac{4}{\sqrt{2}},\frac{3}{\sqrt{2}}\right),\ A:(4\sqrt{2},0),\ B:(0.3\sqrt{2})$

Equation of tangent is $\frac{x\left(\frac{1}{\sqrt{2}}\right)}{4}+\frac{y\left(\frac{1}{\sqrt{2}}\right)}{3}=1$

$\Rightarrow 3x+4y=12\sqrt{2}$

Let $Q:(\alpha,\beta)\Rightarrow 3\alpha+4\beta=12\sqrt{2}$

Equation of $CD: x\alpha+y\beta=12\Rightarrow x\left(\frac{12\sqrt{2}-4\beta}{3}\right)+y\beta=12$

$\Rightarrow(12\sqrt{2}x-36)+\beta(3y-4x)=0$

$\Rightarrow$ Point of concurrency: $\left(\frac{3}{\sqrt{2}},\frac{4}{\sqrt{2}}\right)$

Also, distance of $(0,0)$ from $3x+4y-12\sqrt{2}=0$ is $\frac{12\sqrt{2}}{5}$ as radius of circle is $\sqrt{12}$

$\Rightarrow \dfrac{\text{distance}}{\text{radius}} = \dfrac{\sqrt{24}}{\sqrt{25}} < 1$

$\Rightarrow$ cuts at 2 points.

Let (h,k) be the circumcentre of ΔQCD

$\Rightarrow (h,k)$ is midpoint of $(0,0)$ and (α,β)

$\Rightarrow 2h=\alpha, 2k=\beta$

Also $3\alpha+4\beta=12\sqrt{2} \Rightarrow 3h+4k=6\sqrt{2}$

For R equation of director circle of ellipse

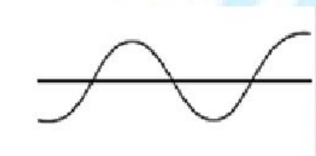
For (iv) equation of director circle of circle

For (S) Any line passing through one focus of ellipse will pass through other focus after reflection.

63-65. (I) $2y^2=2x-1$ *and* $2x^2=2y-1$

Equation of tangents are $y=m\left(x-\dfrac{1}{2}\right)+\dfrac{1}{4m}$ and $y-\dfrac{1}{2}=mx-\dfrac{m^2}{4}$ respectively

So, $-\dfrac{m}{2}+\dfrac{1}{4m}=\dfrac{1}{2}-\dfrac{m^2}{4}$; $m^3-2m^2-2m+1=0$



This equation has 3 real and distinct roots i.e., 3 common tangents

For common normals, they are symmetric about $y=x$

So, only one common normal

(II) After shifting at origin, we get $y^2=\dfrac{4}{\sqrt{3}}x$ *and* $\dfrac{x^2}{9}+\dfrac{y^2}{3}=1$

Equation of normals are $y=mx-\dfrac{2}{\sqrt{3}}m-\dfrac{1}{\sqrt{3}}m^3$ and $y=mx\pm\dfrac{m(a^2-b^2)}{\sqrt{a^2+b^2m^2}}$

So, $\dfrac{2}{\sqrt{3}}m+\dfrac{1}{\sqrt{3}}m^3=\dfrac{6m}{\sqrt{9+3m^2}}$; $m=0, \pm1$. Hence, 3 common normals.

(III) $y=x^2$ and $y=-(x-1)^2$

Equation of tangents are $y=mx-\dfrac{m^2}{4}$ and $y=m(x-1)+\dfrac{m^2}{4}$ respectively

So, we get $m=0,2$. Hence, 2 common tangents

Common normals: Equation are $y=mx+\dfrac{1}{2}+\dfrac{1}{4m^2}$ and $y=mx-m-\dfrac{1}{4m^2}-\dfrac{1}{2}$

So, we get $2m^3+2m^2+1=0$.

Only one real root, so 1 common normal.

(IV) $y^2=-4x$ *and* $x^2=-4y$

Number of common tangents will be same

For $y^2 = -4x \text{ and } x^2 = -4y$, only one common tangent

Common normals: $2m + m^3 = -2 - \frac{1}{m^2}$

$f(m) = m^5 + 2m^3 + 2m^2 + 1$, only one root negative

So, only one common normal.

66. (I) Locus of centre of circle S is parabola

iv) Put $z = x + iy$

$\sqrt{(x-1)^2 + (y-2)^2} = |x - 4 + 1|$

$x^2 - 2x + 1 + y^2 - 4y + 4 = x^2 - 6x + 9$

$\Rightarrow (y-2)^2 = -4x + 8$ represents parabola

67. (II) S touches one circle externally and other circle internally, so centre of S lies on ellipse. So locus is ellipse

(III) $|z - 1 - i| + |z - 1 - 2i| = 4$

Represents ellipse whose foci are (1,1), (1,2) and major axis length 2a=4

68. (III) S touches given two isolated circle externally so locus of centre of S is hyperbola

(ii) $\left||z| - |z - 4 - 5i|\right| = 3$ represents hyperbola whose foci are (0,0), (4,5) and transverse axis length (2a) =3

(IV) Represents line

(i) Represents line

69. y = max touches circle

$\Rightarrow D = 0$ for $(m^2+1)x^2 - 2a(m+2)x + 4a^2 = 0$

$\Rightarrow 4a^2(m+2)^2 = 4(m^2+1)(4a^2)$

$\Rightarrow m = 0, \frac{4}{3}$

Let $A:(x_1, m_1x_1)^2, B:(x_2, m_2x_1)$

$AB^2 = 9 \Rightarrow (x_1 - x_2)^2 + (m_1x_1 - m_2x_2)^2 = 9$ where

$x_1 = \frac{(m_1+2)}{m_1^2+1}a, x_2 \frac{(m_1+2)}{m_2^2+1}a, m_1 = 0, m_2 = \frac{4}{3}$

get $a = \frac{3\sqrt{5}}{4}$

70. Equation of normal at point $T\left(t, \frac{t^2}{2}\right)$ to C $y - \frac{t^2}{2} = \frac{-1}{t}(x - t) \Rightarrow y = \frac{-x}{t} + \frac{t^2}{2} + 1$

If circle S touches curve C at T

$(x(t), y(t))$ must lie on normal

$$\Rightarrow y(t)=\frac{-x(t)}{t}+\frac{t^2}{2}+1$$

Also, distance of centre of circle from T equals radius = distance of centre of circle from x-axis.

$$\Rightarrow (x(t)-t)^2+\left(y(t)-\frac{t^2}{2}\right)^2=(y(t))^2$$

$$x(t)=\frac{t\left(1+\sqrt{t^2+1}\right)}{2}, y(t)=\frac{t^2+1-\sqrt{t^2+1}}{2}$$

$$\Rightarrow x(t)y(t)=\frac{1}{4}t^3\sqrt{t^2+1}$$

$$\lim_{r\to 0^0}\int_r^1 \frac{1}{4}t^3\sqrt{t^2+1}dt$$

Let $\sqrt{1+t^2}=z\Rightarrow 2t\,dt=2zdz$

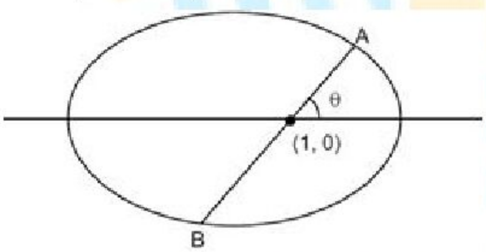
$$\Rightarrow \lim_{r\to 0^-}\int_{\sqrt{1-t^2}}^{\sqrt{2}}\frac{1}{4}\left(z^2-1\right)z^2dz=\frac{\sqrt{2}+1}{30}$$

71. Let AB = I,

Equation of $AB=\frac{x-1}{\cos\theta}=\frac{y-0}{\sin\theta}=r$

$$\Rightarrow \frac{(1+r\cos\theta)^2}{4}+\frac{(r\sin\theta)^2}{3}=1$$

$$\Rightarrow r^2\left(3+\sin^2\theta\right)+6r\cos\theta-9=0$$



$$\Rightarrow r_1-r_2=-\frac{6\cos\theta}{3+\sin^2\theta}, r_1r_2=\frac{9}{3+\sin^2\theta}$$

$$I_1=|r_1|+|r_2|=\sqrt{(r_1-r_2)^2+4r_1r_2}=\frac{12}{3+\sin^2\theta}$$

$$I_2=\frac{12}{3+\sin^2\left(\frac{\pi}{2}+\theta\right)}=\frac{12}{3+\cos^2\theta}$$

$$=\frac{1}{I_1}+\frac{1}{I_2}=\frac{7}{12}$$

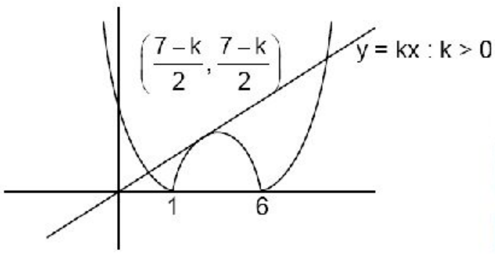
72. normal at $\left(t,t^2\right)$is $2t(y-t)^2=-x+t$

Passes through $\left(0, \frac{3}{2}\right)$.

73. So, $-2x + 7 = k$

$\Rightarrow x = \frac{7-k}{2}$

Also $-\frac{(7-k)^2}{4} - 7\left(\frac{7-k}{2}\right) - 6k = \left(\frac{7-k}{2}\right)$

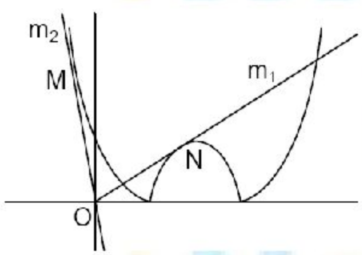


$\Rightarrow k = 7 + \sqrt{24}, 7 - \sqrt{24}$ (Rejected)

$\text{So}, k \in \left[7 - \sqrt{24}, \infty\right)$

Similarly consider case – II If k < 0 then $k \in \left(-\infty, -\left(7 + \sqrt{24}\right)\right]$

Second Method



Clearly: $m_1 < k < k_2$

For slope of ON

$-\left(x^2 - 7x + 6\right) = kx$

Must have equal roots

For slope of OM

$x^2 - 7x + 6 = kx$

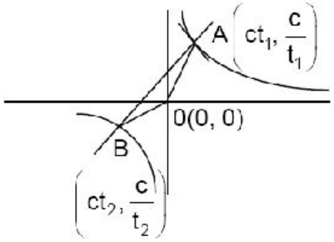
Must have equal roots

74. As $\ell_1^2 + \ell_2^2 = \left(SS_1\right)^2$ (S, S_1 and foci of ellipse)

75. $\Delta PS_1^{'}S_2$ and $\Delta PS_1S_2^{'}$ are congruent

76. Shift the curve so that centre origin, xy = 4

$\because t_2 = \frac{-1}{t_1^3}; m(OA) = \frac{1}{t_1^2}, m(OB) = \frac{1}{t_2^2} = t^6; \tan\theta - \frac{t - t_1^4}{t_1^2}$



Also, $\tan\left(\angle OAB\right) = \dfrac{t_1^2 - \dfrac{1}{t_1^2}}{1+1} = \dfrac{t_1^4 - 1}{2t_1^2}$

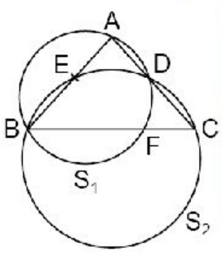
$\therefore \dfrac{\tan\theta}{\tan\left(\Sigma OAB\right)} = -2$

So, $\dfrac{\sin\left(\theta - A\right)}{\sin\left(\theta - A\right)} = 3$

77. Let circumcircle of triangle BDC be S_1 and the circumcircle of triangle ABD be S_2.

So, $\dfrac{AD}{CD} = \dfrac{AB}{BC}.....(1)$

Now, for circle S_1: $AE \times AB = AD \times AC$



$\Rightarrow AE = \dfrac{AD\times}{AB}.....(2)$

For circle S_2: $CF \times CB = CD \times CA$

$\Rightarrow CF = \dfrac{CD \times CA}{CB}.....(3)$

From equation (2) and (3), we get

$\dfrac{AE}{CF} = \dfrac{AD \times CB}{AB \times CD}.....(4)$

From equation (1) and (40), AE = CF

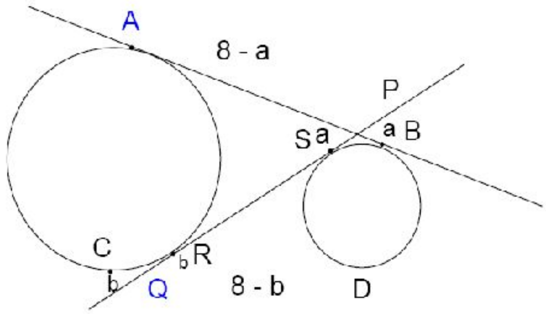
So, CF = 3

78. $R = \dfrac{\dfrac{1}{3}.\dfrac{2}{7}}{1 - \dfrac{1}{3}.\dfrac{2}{7}} = \dfrac{2}{19}$

79. PA = PR = 8 – a

So, PQ = 8 – a + b

Also, QD = QR = 8 – b



So QP = 8 – b + a

$\Rightarrow$ a = b

PQ = 8

80. Intersection point of $x - 2y + 6 = 0$ and $x - y + 2 = 0$ is the incenre of the ΔABC

I (2,4)

r = in radius = Perpendicular distance of I from the line AB

$$= \left| \frac{2 + 12 - 4}{\sqrt{10}} \right|$$

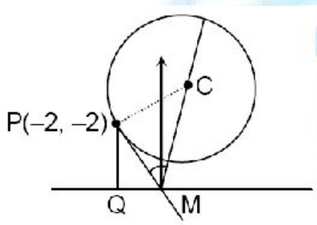
$$r = \sqrt{10}$$

$$\sum \left(r\cot\frac{A}{2} + a \right) = r\cot\frac{A}{2} + a + r\cot\frac{B}{2} + b + r\cot\frac{C}{2} + c$$

$$= s - a + a + s - b + b + s - c + c$$

$$\left[\frac{\Delta}{\Sigma\left(r\cot\frac{A}{2} + a \right)} \right] = \left[\frac{rs}{3s} \right] = \left[\frac{\sqrt{10}}{3} \right] = 1$$

81. $\tan 2\alpha = \frac{PC}{PM} = \frac{5}{PM} ......(1)$



$$\sin(90 - \alpha) = \frac{PQ}{PM}$$

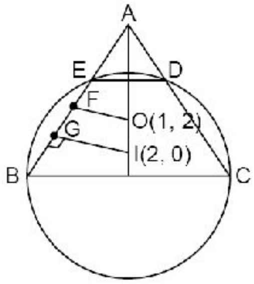
$$PM = \frac{5/2}{\cos\alpha} = \frac{5}{2\cos\alpha} .....(2)$$

$$5\cot 2\alpha = \frac{5}{2\cos\alpha}$$

$$\Rightarrow 10\cot 2\alpha \cos\alpha = 5$$

82. Equation of OI is, $2x + y = 4$

Vertex is point intersection of $2x + y = 4$ and $5x + 2y = 4$



$\therefore A = (-4,12), AO = 5\sqrt{5},\ AI = 6\sqrt{5}$

$\because AG = \dfrac{AE + AB}{2}$

$\therefore AE = 2AG - AB = \sqrt{15}$

$\therefore$ Area of quadrilateral BCDE = area of ΔABC – area of ΔADE

$= \dfrac{\sqrt{3}}{4}\left(AB^2 - AE^2\right) = 90\sqrt{3}$

83. $t_1 + t_2 = 2$

$\dfrac{2 - m_2}{1 + 2m_2} = \sqrt{3}$

and $\dfrac{m_3 - 2}{1 + 2m_3} = \sqrt{3}$

$t_1 + t_2 + t_3 = \dfrac{p}{q} = \dfrac{3}{11}$

84. From similar ΔAEM and ΔAPE

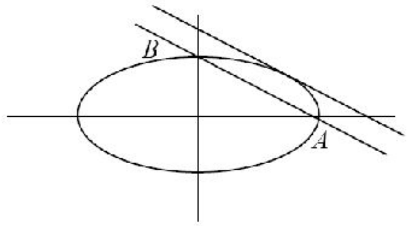
$\dfrac{AE}{AP} = \dfrac{AM}{AE}$

$\Rightarrow AE^2 = AM.AP = (AD - r)AP.....(1)$

As both circles touch each other $(R - r)^2 = r^2 + OD^2 = r^2 + (AD - R)^2$

AE = AD

85. Let perpendicular distance of P from the line b h.



$\frac{1}{2}\times h\times 5=6\left(\sqrt{2}-1\right)\left(\text{as }\Delta PAB=6\left(\sqrt{2}-1\right)\right)$

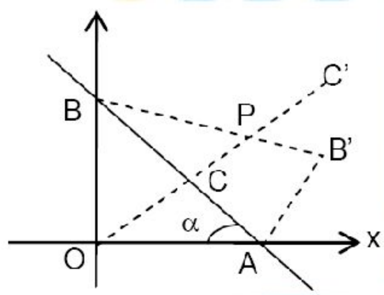
$\Rightarrow h=\frac{12\left(\sqrt{2}-1\right)}{5}$

Now distance of tangent parallel to AB i.e.,

$4y+3x=12\sqrt{2}$, from line AB is $\frac{12\left(\sqrt{2}-1\right)}{5}$

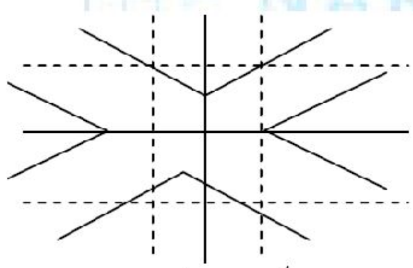
There are just three such points

86. $A\equiv(1,0), B=\equiv(0,\tan\theta), C\equiv\left(\frac{1}{2},\frac{\tan\alpha}{2}\right), B'=\left(\frac{2\tan^2\alpha}{1+\tan^2\alpha},\frac{\tan^3\alpha-\tan\alpha}{1+\tan^2\alpha}\right)$



$\frac{\Delta ABB'}{\Delta BB'C}=2$

87.



$x=\frac{-1}{2}$ $x=\frac{1}{2}$

88. Let $A(t_1), B(t_2)$ and $C(t_3)$ be the points on the parabola

Since $AB \perp BC \Rightarrow \frac{4}{(t_1+t_2)(t_2+t_3)} = -1$

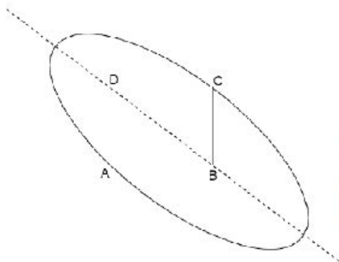
$t_2^2 + t_2(t_1+t_3) + t_1 t_3 + 4 = 0$

$\Rightarrow (t_1+t_3)^2 \geq 4(t_1 t_3 + 4) \Rightarrow (t_1 - t_3)^2 \geq 16$

Also, length of intercept cut off by tangents from y-axis is $|t_1 - t_3| \geq 4$,

Hence 1002.2λ is greater than 4008.80

89.



$(AB)(AD) = 200$

$AB + AD = 2a$

$BD^2 = AB^2 + AD^2 = (2ae)^2$

$(AB+AD)^2 = AB^2 + AD^2 + 2(AB)(AD)$

$\Rightarrow 4a^2 = 4a^2e^2 + 400 \Rightarrow b = 10$

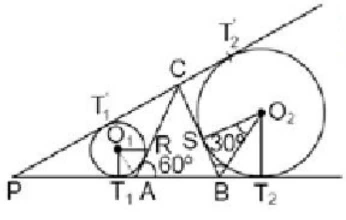
$\Rightarrow a^2(1-e^2) = b^2 = 100 \Rightarrow a = 20$

$P = 2(AB + AD) = 4a = 80$

or $103.44P = 8275.20$

90. $\angle T_1O_1R = 60^0$ and $\angle AO_1R = 30^0$

Similarly, $\angle BO_2S = 30^0$



$T_1T_2 = T_1A + AB + BT_1 = RA + RB + SB$

$= r_1 \tan 30^0 + a + r_2 \tan 30^0 = \frac{r_1 + r_2}{\sqrt{3}} + a$

And $T_1'T_2' = T_1'C + T_2'C = a - RA + a - BS = 2a - \frac{r_1 + r_2}{\sqrt{3}}$

Since common external tangents are equal $T_1T_2 = T_1'T_2'$

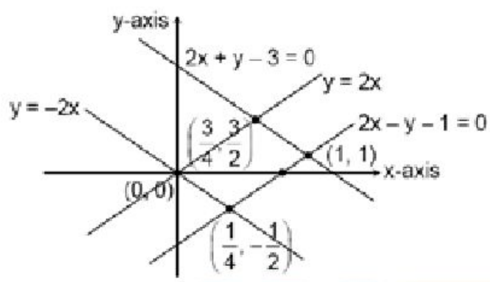
$\Rightarrow r_1 + r_2 = \frac{\sqrt{3}a}{2}$

91. Lines with slope ± 2 and passing through $(0,0)$ and $(1,1)$ are $y = \pm 2x$ and $y - 1 = \pm 2(x-1)$ respectively.

By Mean Value theorem, no point of form $(x, f(x))$ lies outside this parallelogram.

However, we can construct functions which satisfy the given condition and are arbitrarily close to the sides of the parallelogram.

So, $b - a =$ area of parallelogram $= \frac{3}{4}$



92. Let $f(x) = x(4-x)$ and $y = k$

Let A be the region below $y = f(x)$ and above $y = k$

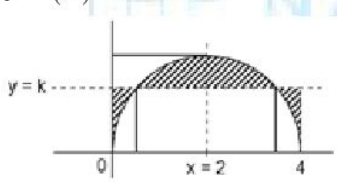
And B be the region below $y = k$ and above $y = f(x)$

then $F(k) = A + B$

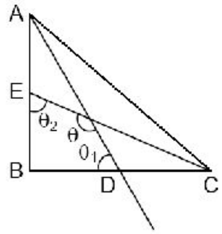
$$A = 2\left(\int_0^k f^{-1}(x)\,dx + \int_k^4 2 - f^{-1}(x)\,dx\right)$$

$$\frac{dA}{dk} = 2\left(f^{-1}(k) - 2 + f^{-1}(k)\right), \frac{dA}{dk} = 0$$

$$f^{-1}(k) = 1 \Rightarrow k = 3$$



93.



$\tan\theta_1 = \frac{AB}{BD} = \frac{2c}{a}$

$\tan\theta_2 = \frac{BC}{BE} = \frac{2a}{c}$

$\tan\theta_1 \cdot \tan\theta_2 = 4$

Angle between median $= \tan^{-1}\left(\frac{3-1}{1+3}\right) = \tan^{-1}\frac{1}{2}$

$\theta = \pi - \tan^{-1}\frac{1}{2}$

Now, $\theta_1 + \theta_2 + \theta = \frac{3\pi}{2}$

$\theta_1 + \theta_2 + \pi - \tan^{-1}\frac{1}{2} = \frac{3\pi}{2}$

$\theta_1 + \theta_2 = \frac{\pi}{2} + \tan^{-1}\frac{1}{2}$

$\tan(\theta_1 + \theta_2) = \tan\left(\frac{\pi}{2} + \tan^{-1}\frac{1}{2}\right) = -\cot\left(\tan^{-1}\frac{1}{2}\right) = -2$

$\frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2} = -2$

$\tan\theta_1 + \tan\theta_2 = -2(1-4) = 6$

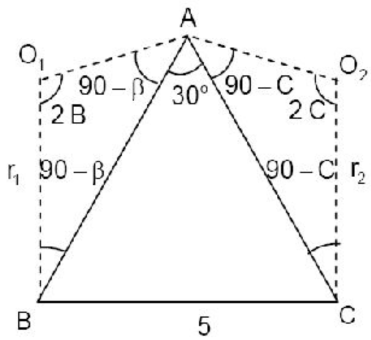
$\frac{2c}{a} + \frac{2a}{c} = 6 \Rightarrow \frac{c^2 + a^2}{ac} = 3 \Rightarrow \frac{64}{ac} = 3 \Rightarrow ac = \frac{64}{3}$

Area $= \frac{1}{2}ac = \frac{32}{3}$

94. Reflection of $(4, 5)$ w.r.t lines $y = x$ and $y = 2x$ are $(5, 4)$ and $\left(\frac{8}{5}, \frac{31}{5}\right)$ respectively.

Minimum value of $PA + PB + AB =$ distance between $(5, 4)$ and $\left(\frac{8}{5}, \frac{31}{5}\right) = \sqrt{16.4}$

95.



In ΔAO_1B, $\dfrac{AB}{\sin 2B} = \dfrac{r_1}{\cos B}$ ...(1)

In ΔAO_2B, $\dfrac{AC}{\sin 2C} = \dfrac{r_2}{\cos C}$ ...(2)

From above equations,

$$\frac{AB.AC}{\sin 2B.\sin 2C} = \frac{r_1 r_2}{\cos B \cos C}$$

$$\frac{2R\sin C.2R\sin B}{2\sin B 2\sin C} = r_1 r_2$$

$$r_1 r_2 = R^2 = \left(\frac{5}{2\sin 30^0}\right)^2 = 25.$$

96. Conceptual

97. Other 2 vertices lie on the director circle

$L = 2\sqrt{5}$

98. If x and y are the sides of two squares then $x^2 + y^2 = 1$

Let without loss of generality $x \ge y$

So, shorter side of the rectangle should be at least x

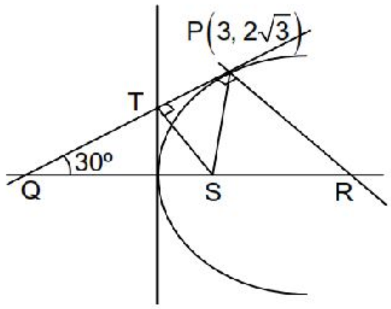
Longer side should be at least $(x + y)$

So, $A \equiv x(x + y)$

Put $x = \cos\theta$, $y = \sin\theta$, $\theta \in \left[0, \dfrac{\pi}{4}\right]$

$A = \dfrac{1}{2} + \dfrac{\sqrt{2}}{2}\cos\left(2\theta - \dfrac{\pi}{4}\right) \le \dfrac{1+\sqrt{2}}{2}$, equality holds if $\theta = \dfrac{\pi}{8}$

99.



Area $PRST = \Delta\ PQR - \Delta\ QST$

$= \frac{1}{2} 8 \times 2\sqrt{3} - \frac{1}{2}\left(\frac{1}{2} \times 4 \times 4 \sin 150^0\right)$

$= 8\sqrt{3} - 2\sqrt{3} = 6\sqrt{3}$

100. $y = x^2$ and $y = -\frac{8}{x}$

$\Rightarrow y_1 = x_1^2$ and $y_2 = -\frac{8}{x^2}$

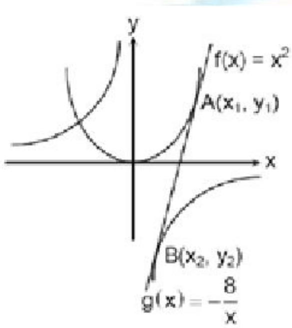
$\because \left(\frac{dy}{dx}\right)_A = \left(\frac{dy}{dx}\right)_B$, we get $2x_1 = \frac{8}{x_2^2}$

$\Rightarrow x_1 x_2^2 = 4$

$m_{AB} \Rightarrow 2x_1 = \frac{x_1^2 + \frac{8}{x_2}}{x_1 - x_2}$

$\Rightarrow x_1 = 4$ and $x_2 = 1,\ y_1 = 16$

$\therefore\ x_1 + y_1 + x_2 = 21$ s

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